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Ambiguous attachment through the Visual World Paradigm

An experimental study of Danish learners of Spanish

Jesús Rabanal Álvarez

Supervised by Alberto Parola

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University of Copenhagen
Faculty of Humanities
MSc in It & Cognition
Emil Holms Kanal 2
2300 Copenhagen S

"A becoming is not a correspondence between relations. But neither is it a resemblance, an imitation, or, at the limit, an identification. Becoming is a verb with a consistency all its own; it does not reduce to, or lead back to, 'appearing,' 'being,' 'equalling,' or 'producing.' "

Deleuze, 1969-1990

"Love does not alter the beloved, it alters itself. The love with which one loves is not the same at every moment; it changes constantly. Yet this change is not unfaithfulness but the very constancy of love, for love is itself the power of transformation. Thus the lover's constancy is not that he remains the same, but that love itself remains in him as the ground of change."

Kierkegaard, 1847-1995

"Love is not love
Which alters when it alteration finds,
Or bends with the remover to remove:
O no! it is an ever-fixed mark
That looks on tempests and is never shaken."

Shakespeare, 1609-2002, Sonnet 116, ll. 2-6

Preface

The process through which I have come to be who I am today can be expressed in the words of three voices that have accompanied me: Deleuze, Kierkegaard, and Shakespeare. I was someone different when I first started this program and when I arrived in Denmark, but through struggle and suffering I have become someone else—shaped by learning, challenge, and the discovery of myself. Deleuze's notion of becoming reflects my own process of continuous change, a movement that never truly stops. Kierkegaard reminds me that change is not something to be feared, but an essential part of being human; we must allow ourselves to be reshaped over time without losing the grounding values that guide us. Finally, Shakespeare's words on love resonate with me as a symbol of persistence: even as life presents obstacles, we must continue forward, for life is worth living and feeling. These voices reflect my own journey over the past two and a half years—a movement between change and constancy, transformation and endurance—that has defined both who I have become and what this journey has meant for me.

Acknowledgements

This thesis I dedicate to my mother. And I would like to express my sincere gratitude to my family and friends for their constant support throughout this journey. In particular, I am deeply thankful to my father, whose unconditional support and unbreakable trust in me have been a continuous source of strength and motivation. His confidence in my abilities has been invaluable at every stage of this work.

I am also grateful to my friends for their encouragement, patience, and understanding during the demanding periods of this research.

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Abstract

Anaphora resolution is a central component of sentence comprehension, requiring comprehenders to establish referential links between pronouns and their potential antecedents. In sentences involving structural ambiguity, pronouns may plausibly refer to more than one antecedent, giving rise to competing high-attachment and low-attachment interpretations. Previous research has shown systematic cross-linguistic differences in attachment preferences, with English speakers generally favoring low attachment, while Romance languages such as Spanish tend to prefer high attachment. However, comparatively little is known about how such ambiguities are resolved by second language learners, particularly when the learners' first language and target language may differ in their attachment preferences.

This thesis investigates how native speakers of Spanish, native speakers of Danish, and Danish learners of Spanish resolve ambiguous pronoun attachment during online sentence processing. Danish constitutes a particularly relevant case, as subject–object attachment ambiguities have received limited empirical attention in the language, and its attachment preferences remain underdocumented. By including both Spanish and Danish native-speaker baselines, the study provides reference points for evaluating whether Danish learners of Spanish converge on the Spanish high-attachment pattern, show influence from Danish, or exhibit a more variable processing profile.

The study used an eye-tracking methodology within a visual-world paradigm. Participants viewed scenes containing two potential antecedents and a scenery/background region while listening to sentences containing ambiguous pronouns. Eye movements were recorded continuously, and fixation allocation to subject- and object-related regions was used as an online measure of attachment preference. Com-

prehension question responses were also collected as an offline measure of final interpretation.

The results showed that native speakers of Spanish exhibited a clear high-attachment preference, reflected in both fixation-based measures and comprehension responses. Native speakers of Danish did not show a reliable attachment preference under the present experimental conditions; instead, fixation allocation and comprehension responses were relatively balanced across subject- and object-related interpretations. Danish learners of Spanish showed an intermediate aggregate pattern, with a weak numerical tendency toward subject-oriented processing, but no statistically reliable group-level preference. Their behavior therefore did not fully align with either the Spanish or Danish baseline.

Overall, the findings suggest that attachment preferences vary across languages and that second language attachment resolution cannot be explained solely in terms of full convergence on the target language or direct transfer from the first language. Instead, the results point toward a more cautious account in which L2 processing may involve partial target-language alignment, weak L1 influence, task-specific processing demands, and unresolved individual variability. More broadly, the thesis contributes new empirical evidence on Danish attachment ambiguity resolution and Danish L2 Spanish processing, while demonstrating the usefulness of combining fixation-based measures with comprehension responses in the study of real-time anaphora resolution.

Contents

1 Introduction

1.1 Human Communication in Real Time

Human communication is a complex cognitive and social phenomenon that unfolds dynamically in real time. Language, as one of its central components, allows individuals to encode, transmit, and interpret information through structured symbolic systems. During everyday interaction, speakers continuously produce linguistic signals while listeners must rapidly decode and interpret them, often under conditions of uncertainty and ambiguity.

Crucially, language comprehension does not occur after an utterance has been fully articulated. Rather, comprehension proceeds incrementally: listeners interpret incoming linguistic input word by word (Isasi-Isasmendi *et al.*, 2024), constructing and updating mental representations as each new element becomes available. This incremental nature of processing means that the human cognitive system must constantly generate predictions, evaluate competing interpretations, and resolve ambiguities in real time i.e.: Figure 1.1.

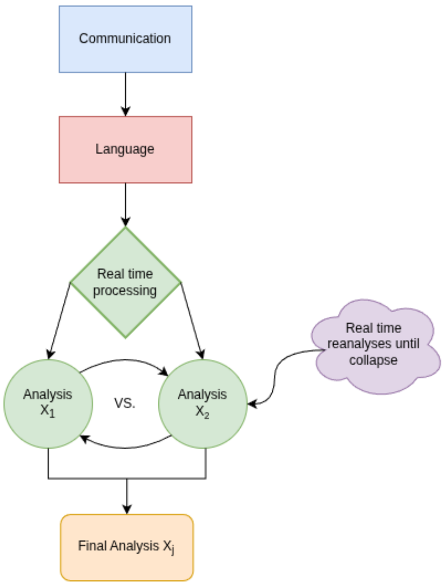


Figure 1.1.: Reanalyses flowchart

One of the most cognitively demanding aspects of real-time language comprehension involves resolving referential expressions—particularly pronouns—whose interpretation depends on identifying the appropriate antecedent within a discourse representation. In cases where multiple potential antecedents are available, listeners must determine which entity the pronoun refers to while the sentence continues to unfold. Such situations provide a powerful window into the mechanisms underlying sentence processing and ambiguity resolution.

This thesis focuses on one specific instance of referential ambiguity: attachment ambiguity in anaphora resolution. Before examining this phenomenon empirically, it is necessary to establish the theoretical foundations that motivate its investigation. The following sections therefore introduce key concepts related to incremental sentence processing, structural ambiguity, and referential interpretation.

1.2 Incrementality in Sentence Processing

One of the defining characteristics of human language comprehension is its incremental nature. Rather than waiting for a complete sentence before constructing meaning, listeners interpret linguistic input as it unfolds over time. Each incoming word is rapidly integrated into an evolving representation of the utterance, allowing comprehenders to continuously update their interpretation of the message being conveyed. This incremental processing enables efficient communication in real-time interaction, where linguistic input is transient and must be interpreted rapidly.

Incremental processing implies that language comprehension is not a purely reactive process, but rather a dynamic and predictive one (Ryskin and Nieuwland, 2023). As listeners receive partial linguistic input, they actively generate expectations about the continuation of the sentence. These expectations can concern syntactic structure, semantic content, and potential referents within the discourse. Predictive mechanisms therefore play an important role in facilitating efficient comprehension, as anticipated structures or meanings can be integrated more easily once they are encountered in the input.

The predictive nature of sentence processing also means that multiple potential interpretations may become activated simultaneously when ambiguity arises. When the linguistic signal allows for more than one possible structural or referential interpretation i.e.: Figure 1.1, the comprehension system must evaluate and select among competing alternatives. Different sources of information—including syntactic structure, semantic plausibility, discourse context, and processing constraints—can influence which interpretation becomes dominant. This process is often described as a form of competition between candidate interpretations that are weighted according to their compatibility with the available evidence.

In addition to predictive mechanisms, theoretical accounts differ in how they conceptualize the interaction between sources of information during incremental processing. Constraint-based approaches (Frazier, 1995) propose that multiple sources of information—syntactic, semantic, and contextual—are integrated simultaneously, with interpretations emerging from the weighted interaction of these constraints. In contrast, more modular or two-stage accounts suggest that initial parsing decisions may rely primarily on syntactic structure, with other sources of information influencing interpretation at a later stage. The extent to which these processes operate in parallel or sequentially remains a central question in the study of sentence comprehension.

Importantly, the incremental architecture of sentence processing means that such interpretive decisions must be made under temporal constraints. Listeners cannot indefinitely delay interpretation until all relevant information has been received. Instead, provisional analyses are constructed and continuously revised as new input becomes available. This property of language processing makes ambiguous constructions particularly informative for the study of comprehension mechanisms (Keith Rayner, 1991). By examining how listeners resolve ambiguity in real time, researchers can gain insight into the strategies and constraints that guide sentence interpretation.

In the context of the present thesis, the incremental nature of sentence processing is particularly relevant because the interpretation of pronouns often depends on information that becomes available progressively within a sentence. When multiple potential antecedents are introduced before a pronoun appears, comprehenders must rapidly evaluate which entity the pronoun most likely refers to. Observing how this process unfolds provides valuable evidence about the mechanisms underlying referential interpretation. To better understand this phenomenon, it is necessary to examine the nature of referential expressions and the processes involved in linking them to discourse entities.

1.3 Referential Expressions and Anaphora

A central component of language comprehension involves establishing reference between linguistic expressions and entities represented in a discourse model. Referential expressions are linguistic forms that allow speakers to introduce, maintain, or reintroduce entities within a conversation (Owen and Chomsky, 1986). These expressions include proper names, definite descriptions, and pronouns, each of which provides different types of information about the intended referent.

Proper names and definite descriptions typically provide relatively rich information about the entity being referred to. In contrast, pronouns are highly underspecified expressions whose interpretation depends heavily on the surrounding linguistic and discourse context. Because pronouns often lack descriptive content, listeners must rely on cues such as syntactic structure, discourse prominence, and accessibility in order to identify the intended referent. This makes pronoun resolution a cognitively demanding process that requires the integration of multiple sources of information (Carreiras and Clifton, 1999).

The process of linking a pronoun to its antecedent is commonly referred to as anaphora resolution. During comprehension, listeners maintain a mental representation of previously mentioned entities in the discourse (Grosz *et al.*, 1995). When a pronoun is encountered, the comprehension system must retrieve a suitable antecedent from this representation. In many cases, the appropriate referent can be identified relatively easily because contextual cues strongly favor one candidate over others. However, situations frequently arise in which multiple potential antecedents are available, creating referential ambiguity.

Several factors have been shown to influence the accessibility of potential antecedents during anaphora resolution (Cuetos and Mitchell, 1988). Syntactic factors such as grammatical role can affect prominence, with subjects often serving as more accessible referents than objects. Discourse-level factors, including topicality and recency, also play an important role in determining which entities are most readily retrieved during processing (Clifton *et al.*, 2007). Additionally, cognitive constraints related to memory and attention may influence the likelihood that a particular discourse entity will be selected as the antecedent of a pronoun (Frazier, 1995).

From a discourse perspective, theories such as Centering Theory (Grosz *et al.*, 1995) have emphasized the role of coherence and attentional state in guiding referential interpretation. According to this view, certain entities within a discourse are more central or prominent than others, for example, a proper noun such as “Mr. Smith” may be more central than a common noun such as “the fountain”, and pronouns tend to refer to these highly accessible entities. This notion aligns with accessibility-based accounts (Schmitz *et al.*, 2024), which propose that referents vary in their activation level within memory, influencing the ease with which they can be retrieved during comprehension.

Because pronouns rely so heavily on contextual information for their interpretation, they provide a valuable window into the interaction between syntactic structure, discourse representation, and real-time processing mechanisms. When multiple structurally valid antecedents are present, listeners must rely on cues from several levels of linguistic representation to resolve the ambiguity (Keith Rayner, 1991). In such cases, referential interpretation intersects closely with structural properties of the sentence, particularly when competing syntactic structures license different antecedent candidates. This brings us to the broader phenomenon of structural ambiguity in sentence comprehension.

1.4 Structural Ambiguity in Language Comprehension

Ambiguity is an inherent property of natural language. Linguistic expressions often permit more than one possible interpretation, requiring listeners to determine which meaning is intended based on contextual and structural cues. Ambiguities can arise at several levels of linguistic representation, including lexical ambiguity, semantic scope ambiguity, and structural ambiguity. Among these, structural ambiguity has played a particularly important role in the study of sentence processing.

Structural ambiguity occurs when a sequence of words can be associated with more than one syntactic structure. Because sentences are organized hierarchically rather than linearly, different structural configurations can lead to different interpretations of the same string of words. During comprehension, the parsing system (Frazier and Fodor, 1978) must determine which hierarchical structure best represents the intended meaning of the sentence. When multiple syntactic analyses are possible, the parser must rely on various constraints to select the most plausible structure.

Research in sentence processing has proposed several strategies that may guide the construction of syntactic structure during comprehension. Early models of parsing suggested that the comprehension system follows relatively simple structural heuristics when building syntactic representations (Zhang, 2020). For example, principles such as minimal attachment (Frazier and Fodor, 1978) and late closure (Igoa *et al.*, 1998) have been proposed to explain how listeners initially favor certain structural analyses over others. However, subsequent research (Maia *et al.*, 2007) has demonstrated that parsing decisions are influenced by a wider range of factors, including lexical information, semantic plausibility, discourse context, and probabilistic expectations derived from language experience.

Structural ambiguity is particularly useful for investigating how these different sources of information interact during comprehension. When multiple syntactic structures are possible, the interpretation selected by the listener can reveal the relative influence of structural preferences, semantic cues, and contextual information. Experimental methods such as eye-tracking have been widely used to examine these processes, as they allow researchers to observe the temporal dynamics of interpretation as listeners encounter ambiguous constructions.

A well-documented consequence of structural ambiguity is the occurrence of garden-path effects, where an initially preferred interpretation must be revised in light of disambiguating information. Such reanalysis processes highlight the dynamic nature of sentence comprehension, as listeners may commit to an interpretation early on and subsequently adjust it when it becomes incompatible with the incoming input. The cognitive cost associated with reanalysis provides important evidence about the mechanisms and constraints involved in parsing (Cunnings *et al.*, 2017).

Within the broader domain of structural ambiguity, attachment ambiguities represent a specific and well-studied phenomenon. These ambiguities arise when a particular phrase or element of a sentence can be attached to more than one potential position within the syntactic structure. In such cases, different attachment decisions lead to different interpretations of the sentence. The study of attachment ambiguities has provided important insights into how syntactic structure and discourse information jointly influence language comprehension. The following section focuses on attachment ambiguity in greater detail, with particular attention to how it interacts with the resolution of ambiguous pronouns.

1.5 Eye-Tracking as a Method for Studying Language Processing

Investigating how language is processed in real time requires methodologies that can capture the temporal dynamics of comprehension. One of the most widely used techniques in psycholinguistics is eye-tracking, which provides a fine-grained, continuous measure of visual attention during language processing (Dussias, 2010). By recording participants' eye movements while they read or listen to linguistic input, researchers can obtain detailed information about how and when different elements of a sentence are processed.

The underlying assumption of eye-tracking is that eye movements are closely linked to cognitive processes. In reading tasks, where visual input corresponds directly to linguistic material, this relationship is particularly robust. In visual-world paradigms, where participants listen to spoken language while viewing a visual scene, eye movements reflect the allocation of attention to potential referents in the environment (Cunnings *et al.*, 2017). In both cases, patterns of fixations and saccades provide insight into the processes involved in comprehension.

Eye-tracking offers several advantages over other experimental methods. Unlike offline measures such as comprehension questions or acceptability judgments, eye-tracking captures processing as it unfolds in real time. This makes it possible to detect moment-by-moment changes in interpretation, including the activation of competing representations and the resolution of ambiguity. As such, eye-tracking has become a central tool in the study of incremental sentence processing and referential interpretation.

1.6 Eye Movements as Indicators of Cognitive Processing

Eye movements during language comprehension are typically analyzed in terms of fixations and saccades. Fixations correspond to periods during which the eye remains relatively stable and visual information is processed, while saccades are rapid movements between fixation points. The duration, location, and sequence of fixations provide valuable information about the allocation of attention and the underlying cognitive processes (Cullipher *et al.*, 2018).

In reading studies, longer fixation durations are often interpreted as reflecting increased processing difficulty. For example, words that are syntactically complex, semantically unexpected, or ambiguous tend to elicit longer fixations or additional regressions (Clifton *et al.*, 2007). Regressive eye movements, in which the reader looks back to earlier parts of the sentence, are commonly associated with reanalysis processes and the resolution of ambiguity.

In visual-world paradigms, fixations are used to infer how listeners map linguistic input onto potential referents in a visual scene. When multiple candidate referents are present, the proportion of fixations directed toward each object can be used to track the activation and competition between different interpretations. Changes in fixation patterns over time provide a dynamic measure of how listeners resolve referential ambiguity as the sentence unfolds.

Importantly, eye movement data must be interpreted within a temporal framework. Because there is a delay between linguistic input and eye movement responses, analyses typically consider time windows aligned with critical regions of the stimulus (Cullipher *et al.*, 2018). This allows researchers to relate patterns of visual attention to specific stages of linguistic processing.

1.7 Eye-Tracking and the Study of Ambiguity Resolution

Eye-tracking has played a crucial role in advancing our understanding of how ambiguity is resolved during language comprehension. By providing continuous, time-sensitive measures of processing, eye-tracking allows researchers to observe how competing interpretations are activated, maintained, and eventually resolved.

In the context of structural ambiguity, eye-tracking studies have shown that listeners often consider multiple interpretations in parallel, particularly in the early stages of processing. Fixation patterns can reveal whether alternative structures are simultaneously entertained or whether one interpretation is initially preferred and later revised. These findings have contributed to ongoing debates regarding the nature of parsing strategies and the role of predictive processing (Cuetos and Mitchell, 1988).

With respect to referential ambiguity, eye-tracking has been especially informative in visual-world paradigms. When a pronoun is encountered, shifts in visual attention toward different potential antecedents can be observed in real time. The proportion of fixations directed at each referent provides a quantitative measure of their relative activation, allowing researchers to track how listeners resolve ambiguity dynamically.

In the present thesis, eye-tracking is used as the primary methodological tool to investigate attachment ambiguity in anaphora resolution. By analyzing fixation patterns across different regions of interest, it is possible to examine how participants allocate attention to competing referents and how this allocation evolves over time. This approach provides a direct window into the interaction between structural and referential processes during real-time language comprehension.

Theoretical Background

2

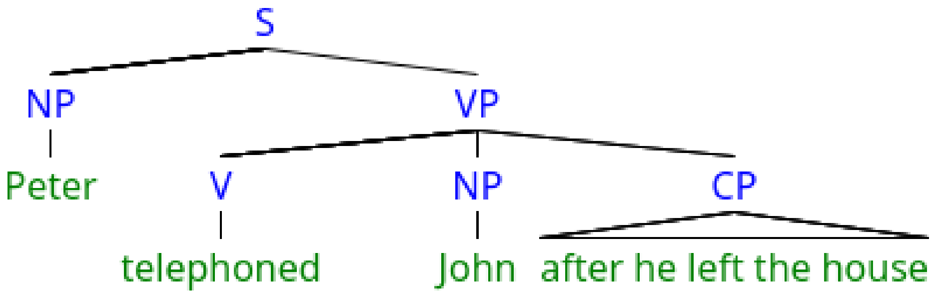
2.1 Attachment Ambiguity and Cross-Linguistic Variation

Attachment ambiguity constitutes a central phenomenon in the study of sentence processing, particularly at the interface between syntax and discourse interpretation. As introduced in the previous section, attachment ambiguities arise when a constituent—such as a subordinate clause—can be structurally integrated at more than one position within a sentence. These alternative attachment sites correspond to distinct hierarchical representations, which in turn license different interpretations of referential expressions such as pronouns.

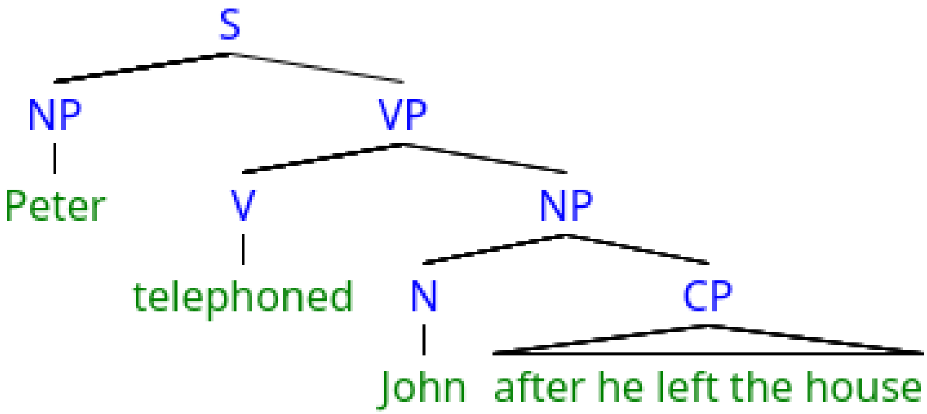
In the case of anaphora resolution, attachment ambiguity directly affects the set of possible antecedents available for interpretation. As illustrated in Figures 2.1a and 2.1b, a subordinate clause may attach either to the object noun phrase (low attachment) or higher in the syntactic structure (high attachment). These structural differences determine whether the pronoun is more naturally interpreted as referring to the object or the subject, respectively. Consequently, attachment decisions and referential interpretation are tightly intertwined processes, making this type of ambiguity particularly informative for investigating real-time language comprehension.

2.1.1 High and Low Attachment Preferences

A substantial body of psycholinguistic research has demonstrated that language users exhibit systematic preferences when resolving attachment ambiguities. Early work by Frazier and Fodor (1978) proposed that the human sentence parser follows general structural principles, such as Late Closure, which favor attaching incoming material to the most recently processed phrase. Under this account, low



(a) High-attachment interpretation.



(b) Low-attachment interpretation.

Figure 2.1.: Schematic representation of high- and low-attachment structures.

attachment should be preferred across languages, as it represents the structurally simpler and more local integration.

Empirical evidence from English initially supported this prediction, with speakers showing a consistent preference for low attachment interpretations in ambiguous sentences. However, subsequent cross-linguistic research revealed that this preference is not universal. In particular, studies on Spanish by Cuetos and Mitchell (1988) and Carreiras and Clifton (1999) demonstrated a robust tendency toward high attachment, in contrast to the low attachment bias observed in English. Similar high attachment preferences have also been reported for other Romance languages, although the strength of the effect may vary across languages and constructions.

These findings challenged strictly universal accounts of parsing and suggested that attachment preferences may be shaped by language-specific factors. As a result, the study of attachment ambiguity has become a key testing ground for theories of sentence processing, particularly with respect to the relative roles of structural principles and experience-based influences.

2.1.2 Theoretical Accounts of Attachment Preferences

Two broad classes of explanations have been proposed to account for cross-linguistic differences in attachment preferences. The first class emphasizes universal parsing strategies, according to which the human sentence processing system operates according to general structural principles that are largely independent of the specific language being processed. Within this framework, apparent cross-linguistic differences are often attributed to methodological factors, lexical biases, or the influence of contextual information rather than fundamental differences in parsing mechanisms.

In contrast, the second class of explanations argues for language-specific tuning of the parser. According to the tuning hypothesis, attachment preferences are shaped by exposure to the statistical regularities of a given language. Under this view, comprehenders learn from experience which structural configurations are more frequent or more likely in their language, and these learned expectations guide parsing decisions during comprehension. This account provides a natural explanation for why speakers of different languages exhibit different attachment preferences, as these preferences reflect differences in input distribution.

More recent approaches have adopted probabilistic and constraint-based frameworks, in which multiple sources of information—including syntactic structure, lexical properties, semantic plausibility, and discourse context—are integrated during processing. Within these models, attachment preferences emerge from the interaction of competing constraints, rather than being determined by a single structural principle. This perspective aligns with broader views of language comprehension as a predictive and experience-driven process, as discussed in Section 1.

2.1.3 Cross-Linguistic Variation in Attachment

Cross-linguistic research has consistently demonstrated that attachment preferences vary across languages. As noted above, English speakers tend to favor low attachment, whereas Spanish speakers typically show a preference for high attachment (Cuetos and Mitchell, 1988; Carreiras and Clifton, 1993). Dutch has been reported to exhibit more variable patterns, with some studies suggesting a weaker or context-dependent preference (Brysbaert and Mitchell, 1996). Italian, similarly to Spanish, has often been associated with a high attachment bias, although variation across experimental conditions has also been observed.

Several factors have been proposed to account for these differences. One line of explanation focuses on syntactic and morphological properties of individual languages, which may influence the availability or salience of particular attachment sites. Another line of research highlights the role of prosody, suggesting that implicit prosodic phrasing during silent reading may bias comprehenders toward one interpretation over another. Additionally, frequency-based accounts emphasize the role of distributional patterns in the input, whereby repeated exposure to certain constructions reinforces particular parsing strategies.

Importantly, cross-linguistic variation in attachment preferences provides strong evidence against strictly universal parsing models. Instead, it suggests that the sentence processing system is flexible and sensitive to language-specific experience. This has significant implications for the study of second language acquisition, where learners must adapt to potentially different parsing preferences in the target language.

2.1.4 Attachment Ambiguity in Second Language Processing

The question of how second language (L2) learners resolve attachment ambiguities has been the subject of considerable debate. One central issue concerns whether L2 learners are able to acquire native-like parsing strategies or whether they rely primarily on strategies transferred from their first language (L1). Empirical findings have been mixed, with some studies showing convergence toward native-like preferences and others indicating persistent L1 influence.

According to transfer-based accounts, L2 learners initially apply the parsing strategies of their L1 when processing the target language. In the context of attachment ambiguity, this would predict that learners whose L1 favors low attachment would also show a low attachment bias when processing an L2 that favors high attachment, at least in the early stages of acquisition. Over time, increased exposure to the L2 may lead to a gradual shift toward target-like preferences, although this process may be incomplete or influenced by proficiency and task demands.

Alternative accounts propose that L2 learners rely more heavily on general cognitive strategies and may exhibit less stable or more variable parsing behavior compared to native speakers (Clahsen and Felser, 2006). Within this framework, learners may show sensitivity to both L1 and L2 cues, resulting in hybrid patterns of interpretation. This variability is particularly evident in online measures, where moment-by-moment processing can reveal competition between alternative interpretations.

2.1.5 Relevance for the Present Study

The present thesis builds on this body of research by examining attachment ambiguity in a previously underexplored population: Danish learners of Spanish. Given the cross-linguistic differences observed between Germanic and Romance languages, this population provides an ideal testing ground for investigating the interaction between L1 transfer and L2-specific parsing strategies.

By employing eye-tracking methodology, the study captures the temporal dynamics of attachment decisions as they unfold in real time. Rather than relying solely on offline judgments, the analysis focuses on fixation patterns as an index of attentional allocation and interpretive preference. This approach allows for a more fine-grained examination of how competing antecedents are evaluated during processing, and whether learners exhibit native-like high attachment preferences or retain L1-based strategies.

In this way, the study contributes to ongoing debates on the nature of sentence processing mechanisms, the role of experience in shaping parsing preferences, and the extent to which these preferences can be modified through second language acquisition.

3 Rationale & Research Gap

The resolution of referential ambiguity is a central problem in sentence processing, as language users must rapidly determine how pronouns and other anaphoric expressions map onto potential antecedents in real time (Clifton *et al.*, 2007). In cases of structural ambiguity involving competing subject and object antecedents, such as those arising in attachment ambiguities, comprehenders are required to evaluate multiple syntactically possible interpretations. These situations are particularly informative for psycholinguistic research because they reveal how the parser selects, maintains, or revises interpretations while linguistic input unfolds.

Decades of research have shown that attachment decisions are not random, but instead reflect processing preferences that may vary across languages (Carreiras and Clifton, 1993). Such preferences have been taken as evidence that sentence processing is shaped by an interaction between general parsing mechanisms and language-specific grammatical, discourse, and probabilistic constraints. Attachment ambiguity therefore provides a useful testing ground for examining whether sentence processing is governed primarily by universal parsing principles or whether parsing strategies are adapted to the distributional and grammatical properties of individual languages.

A well-established finding in this literature is the cross-linguistic contrast between low-attachment preferences reported in English and high-attachment preferences reported for Romance languages such as Spanish and Italian (Cuetos and Mitchell, 1988; Carminati, 2002). While early parsing models proposed universal principles such as minimal attachment and late closure, subsequent work demonstrated that attachment strategies differ across languages, challenging strong universalist accounts (Cunnings *et al.*, 2017). This shift has motivated a large body of research examining how syntactic structure, discourse prominence, referential accessibility, and language-specific experience jointly guide anaphora resolution during online processing.

Despite extensive work on native speakers of languages such as English, Spanish, Italian, and Dutch, Danish remains underrepresented in this line of research. This is a significant gap because Danish provides an important test case for evaluating whether attachment preferences reported for other Germanic languages generalize more broadly. Assuming that Danish patterns like English without direct empirical evidence would be problematic, especially given that attachment preferences may be sensitive to language-specific syntactic and discourse-pragmatic properties. A Danish baseline is therefore necessary in order to establish how native speakers of Danish resolve this type of ambiguity under the present experimental conditions.

Considerably less is also known about how second language (L2) learners resolve attachment ambiguities, particularly when the relevant attachment preferences may differ between the first language (L1) and the target language. Research in L2 sentence processing has shown that learners may converge on native-like parsing strategies, rely on L1-based heuristics, or exhibit intermediate and unstable patterns that vary with proficiency, exposure, and task demands (Contemori *et al.*, 2019; Clahsen and Felser, 2006). However, empirical evidence remains mixed, and relatively few studies have examined attachment ambiguity resolution using online measures in L2 populations. Danish learners of Spanish therefore constitute an especially relevant group, as they allow the relationship between a less-studied L1 baseline and a well-documented Spanish target-language pattern to be examined directly.

The present thesis also builds on previous work using similar experimental materials and procedures to investigate attachment ambiguity resolution. In particular, the design is informed by previous research on pronoun resolution and L2 processing, including work by Cunnings and colleagues (Cunnings *et al.*, 2017), as well as by the materials and methodological experience developed in my previous bachelor thesis. This continuity is important because it allows the present study to adapt a tested experimental logic to a new language pairing and a new learner population. Rather than developing an entirely new paradigm from scratch, the thesis extends an existing research framework to address a specific empirical gap: the processing of attachment ambiguity in Danish and Danish L2 Spanish.

Methodologically, much existing work on anaphora resolution has relied on offline measures such as acceptability judgments, interpretation choices, or questionnaire-based tasks. While informative, these approaches provide limited insight into the temporal dynamics of processing and may obscure intermediate states of competition between interpretations. Offline responses reveal which interpretation participants ultimately select, but they do not necessarily show how competing antecedents were considered during comprehension. This distinction is central to the present thesis, since attachment ambiguity resolution may involve temporary competition between referents even when the final interpretation appears categorical.

Eye-tracking methodology offers a powerful complement by allowing researchers to observe how visual attention is allocated as comprehension unfolds. In a visual-world paradigm, participants listen to linguistic input while viewing a scene containing potential referents. Their eye movements can then be used to examine how attention is distributed across relevant visual regions during sentence processing. Fixation counts and fixation proportions provide graded measures of attentional allocation, making eye tracking particularly well suited for investigating ambiguity resolution in real time.

The present thesis adopts an eye-tracking approach within a visual-world paradigm to investigate how attachment preferences manifest during online sentence processing. By presenting participants with visual scenes containing multiple potential antecedents and tracking their gaze behavior while they process ambiguous sentences, the study operationalizes attachment preferences in terms of fixation-based measures rather than relying only on explicit interpretive judgments. This approach makes it possible to examine not only which interpretation is selected in comprehension responses, but also how attention is distributed across competing referents during processing.

The rationale for this approach is twofold. First, it allows for a more fine-grained analysis of attachment preferences by capturing graded differences in attentional allocation rather than only categorical interpretation outcomes. Second, it provides a principled way to compare native speakers and L2 learners using the same operational definitions and measures, thereby reducing confounds related to task format or metalinguistic reflection. By combining aggregate fixation measures with comprehension responses, the study compares online attentional patterns with offline interpretation choices across the three participant groups.

Overall, this thesis aims to contribute to ongoing debates on cross-linguistic variation and transfer in sentence processing by providing new empirical evidence from Spanish, Danish, and Danish learners of Spanish using online eye-tracking measures. In doing so, it bridges research on anaphora resolution, attachment preferences, and second language processing, while addressing a specific empirical gap concerning Danish attachment ambiguity resolution and its role in L2 Spanish processing.

4 Research Questions and Hypotheses

This study examines how attachment ambiguities in anaphora resolution are processed across different language backgrounds. Building on previous research on cross-linguistic variation in attachment preferences and real-time sentence processing, it explores whether such preferences are maintained, transferred, or restructured in a second language context. In particular, the comparison between native speakers of Spanish, native speakers of Danish, and Danish learners of Spanish allows for an examination of both baseline processing patterns and cross-linguistic influence.

A central motivation for this work lies in the observation that attachment preferences differ across languages, with Romance languages such as Spanish typically showing a high-attachment preference, while several Germanic languages have often been associated with low-attachment tendencies. However, Danish remains underrepresented in this line of research, and little is known about how Danish speakers resolve this type of ambiguity, either in their first language or when processing a second language such as Spanish. This makes the Danish baseline an important component of the study rather than a purely confirmatory comparison.

In addition, previous work on second language sentence processing suggests that learners may either converge on target-like parsing strategies or exhibit persistent influence from their first language, depending on factors such as proficiency, exposure, and task demands. Danish learners of Spanish therefore provide a useful test case for examining whether attachment preferences in an L2 align with the target language, remain influenced by the L1, or result in more variable processing patterns.

The use of eye-tracking within a visual-world paradigm allows these questions to be examined through online measures of attention rather than relying only on explicit post-sentence judgments. This is important because attachment preferences may emerge during processing even when final comprehension responses are less categorical, or conversely, explicit responses may obscure uncertainty present during online interpretation. The combination of fixation-based measures and comprehension responses therefore provides a way to compare real-time processing tendencies with final interpretation choices.

Against this background, the following research questions are addressed.

4.1 Research Questions

- RQ1. What attachment preference do native speakers of Spanish exhibit when processing ambiguous pronoun reference in the present experimental materials?

- RQ2. What attachment preference do native speakers of Danish exhibit under comparable experimental conditions?

- RQ3. How do Danish learners of Spanish resolve attachment ambiguities in Spanish, and to what extent does their processing reflect convergence on Spanish attachment preferences, influence from Danish, or a more variable processing pattern?

These research questions aim to establish both native-speaker baselines and cross-linguistic comparisons. RQ1 and RQ2 identify the attachment preferences exhibited by Spanish and Danish speakers under the specific experimental conditions employed here. Establishing such baselines is essential, as attachment preferences may vary depending on the materials, task design, and language-specific processing constraints. RQ3 extends this comparison by examining how Danish learners of Spanish process attachment ambiguities in the target language, thereby enabling an assessment of convergence toward native-like processing, possible L1 influence, or instability in L2 attachment resolution.

The Spanish baseline is necessary because it provides the target-language reference point for the learner group. If native Spanish speakers show the expected high-attachment preference in the present materials, then learner behavior can be meaningfully evaluated in relation to that pattern. Without this baseline, it would be difficult to determine whether Danish learners of Spanish diverge from native Spanish processing or whether the materials themselves fail to elicit a clear Spanish preference.

The Danish baseline is equally important because it provides the L1 reference point for the learner group. Since Danish attachment preferences are not well established in the literature, assuming a Danish pattern based only on findings from other Germanic languages would be problematic. Including native Danish speakers therefore allows the study to examine Danish attachment behavior directly and to evaluate possible L1 influence in the learner group more cautiously.

The learner group connects the two baselines by testing how attachment ambiguity is resolved when the language being processed is Spanish but the first language background is Danish. If learners pattern with native Spanish speakers, this would suggest convergence toward target-language processing strategies. If they pattern with Danish speakers, this would suggest L1 influence. If they fall between the two baselines or show no stable pattern, this would suggest that L2 attachment resolution may involve partial convergence, weak transfer, or variability related to proficiency and processing demands.

4.2 Hypotheses

The following hypotheses are derived from previous findings on attachment preferences and second language sentence processing. Prior research has shown that native speakers of Spanish tend to favor high-attachment interpretations (Carreiras and Clifton, 1993) , whereas several Germanic languages, including English, have been associated with low-attachment preferences (Carreiras and Clifton, 1999). Since Danish has not been extensively studied in this domain, the Danish prediction is exploratory and based on the broader cross-linguistic pattern reported for related Germanic languages.

In second language acquisition contexts, competing accounts predict either convergence toward target-language preferences or the persistence of L1-based processing strategies. A third possibility is that learners show variable or intermediate patterns, particularly when the relevant attachment preferences are weak, unstable, or not fully established in either the L1 or L2.

- H1. Native speakers of Spanish will show a high-attachment preference, operationalized as increased fixation allocation to subject-related regions.

- H2. Native speakers of Danish will show a low-attachment preference, operationalized as increased fixation allocation to object-related regions. Given the limited previous research on Danish attachment preferences, this hypothesis is exploratory and based on patterns reported for other Germanic languages.

- H3a. Danish learners of Spanish will exhibit fixation patterns consistent with the Spanish high-attachment preference, indicating convergence on L2 processing strategies.

- H3b. Alternatively, Danish learners of Spanish will exhibit fixation patterns consistent with Danish attachment preferences or show no stable bias, indicating possible L1 influence, processing variability, or incomplete convergence on the target-language pattern.

These hypotheses are intentionally framed to distinguish between baseline effects and cross-linguistic effects. H1 and H2 concern native-speaker processing in Spanish and Danish, respectively. H3a and H3b concern the learner group and therefore depend on the relationship between the two baselines. In this sense, the interpretation of the L2 results is not independent of the baseline experiments: learner behavior can only be interpreted meaningfully once the Spanish and Danish reference patterns have been established.

The hypotheses are also operationalized in terms of fixation allocation rather than only comprehension responses. This is because the central concern of the thesis is real-time processing, and fixation patterns provide an online measure of how attention is distributed between potential antecedents as the sentence unfolds. Comprehension responses remain important, but they are treated as complementary offline evidence rather than the sole basis for determining attachment preference.

Taken together, these hypotheses provide a framework for evaluating both language-specific attachment preferences and the extent to which such preferences can be reshaped through second language acquisition. By combining native-speaker baselines with learner data, the analysis offers insight into the interaction between structural parsing strategies, cross-linguistic influence, and real-time sentence processing mechanisms.

5 Methods

This study consists of three related eye-tracking experiments designed to investigate how attachment ambiguities involving pronoun reference are resolved during real-time sentence processing. All three experiments employed the same visual-world paradigm, comparable materials, and the same experimental procedure; they differed primarily in participant group and language background. This design allows for direct cross-group comparison while maintaining experimental control over stimuli and task demands.

Experiment 1 established a baseline attachment preference in native speakers of Spanish. Experiment 2 established a baseline for native speakers of Danish. Experiment 3 examined attachment resolution in Danish learners of Spanish, allowing direct comparison with both baseline groups in order to assess second language (L2) convergence and potential first language (L1) transfer effects. By including both native-speaker baselines and a learner group, the study adopts a comparative approach that enables a more fine-grained interpretation of cross-linguistic processing patterns.

Across experiments, participants viewed visual scenes depicting two potential antecedents while listening to recorded sentences containing subject–object attachment ambiguities. Eye movements were recorded continuously throughout the task to capture online measures of visual attention associated with attachment resolution. The use of the visual-world paradigm makes it possible to link linguistic input to referential interpretation in real time, as participants' gaze behavior reflects the moment-by-moment allocation of attention to competing referents.

A total of 16 experimental sentences were presented, together with 24 filler sentences and three training items. Each trial consisted of a familiarization phase (see Figure 5.1), followed by a fixation-contingent onset and the simultaneous presentation of visual and auditory stimuli. Figure 5.2 provides an overview of the experimental workflow.

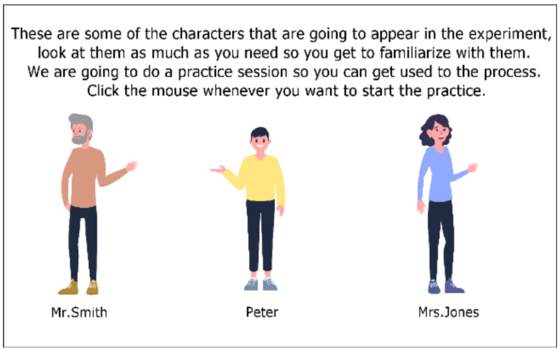


Figure 5.1.: Character familiarization

During the familiarization phase, participants were introduced to the characters appearing in the stimuli in order to ensure a consistent mapping between character names and visual referents across trials. This step was included to minimize variability in referent identification and to ensure that gaze patterns reflected processing decisions rather than uncertainty about character identity.

5.1 Participants

Three participant groups were recruited corresponding to the three experiments. Experiment 1 included native speakers of Spanish, who served as the baseline group for attachment preferences in the target language. Experiment 2 included native speakers of Danish, providing a baseline for attachment preferences in the participants' first language. Experiment 3 included native speakers of Danish learning Spanish as a second language.

Group	Experiment	n	Native language	Spanish proficiency
Spanish baseline	Experiment 1	10	Spanish	Native
Danish baseline	Experiment 2	10	Danish	–
Danish L2 Spanish	Experiment 3	6	Danish	Approx. B1; range A1–C2

Table 5.1.: Overview of participant groups included in the final analyses.

Recruitment was carried out primarily through word of mouth. In addition, flyers and posters were distributed at the University of Copenhagen South Campus and at the University of Granada, Faculty of Philosophy and Letters. Recruitment posts were also shared through social media platforms, including Facebook. Despite these efforts, recruitment for the Danish L2 Spanish group was limited, resulting in a smaller and more heterogeneous learner sample than originally intended.

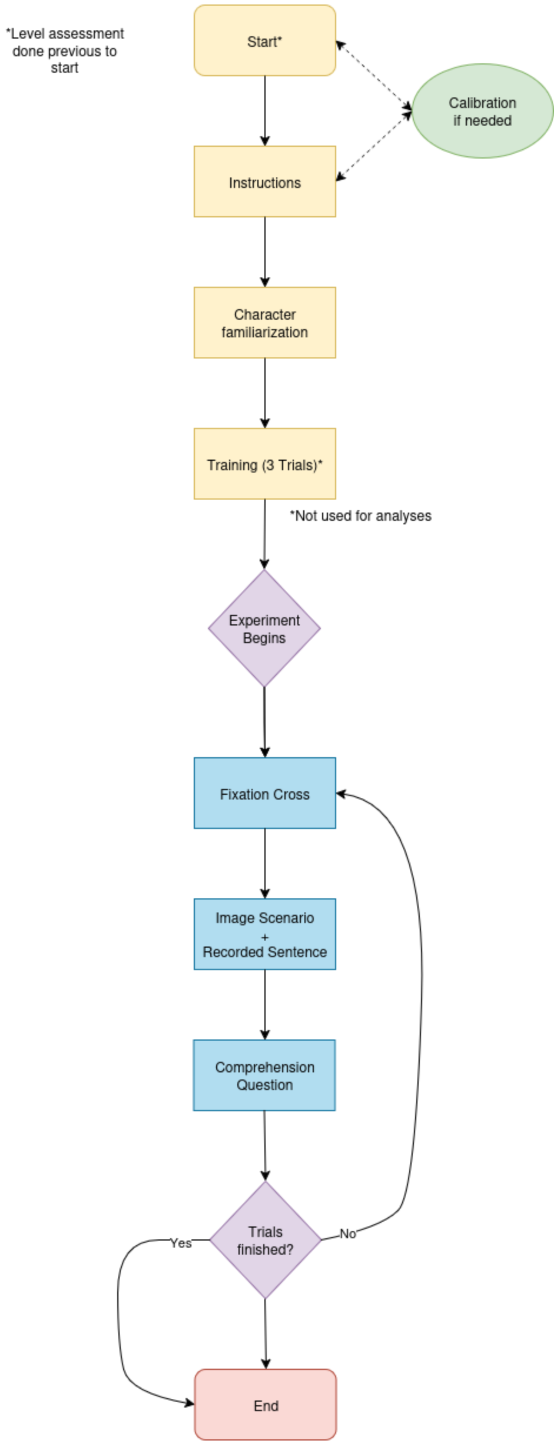


Figure 5.2.: Experimental Workflow

The target was at least ten participants per group. This target was met for both baseline groups (Spanish baseline: $n = 10$; Danish baseline: $n = 10$). For the cross-linguistic experiment, six Danish learners of Spanish were recruited ($n = 6$). While the learner sample is smaller than the two baseline groups, it is treated as an exploratory L2 sample and interpreted with caution throughout the analysis.

Danish learners were intermediate to advanced speakers of Spanish. Language proficiency was assessed using a self-report questionnaire and, where available, supplemented by standardized placement tests. Participants also reported their age of acquisition, length of exposure to Spanish, and frequency of use. These measures were collected in order to characterize the learner group and to provide contextual information for interpreting the L2 results.

All participants included in the final analyses reported normal or corrected-to-normal vision and no history of neurological or language-related disorders. Participants provided informed consent prior to participation.

5.2 Materials

The experimental materials consisted of recorded sentences containing ambiguous pronoun reference, paired with visual scenes depicting two potential antecedents corresponding to the subject and the direct object of the main clause. Critical sentences involved temporal subordinate clauses containing a pronoun whose antecedent was ambiguous, allowing both high-attachment and low-attachment interpretations.

For example, a sentence such as *Pedro llamó a Juan después de que él salió de la casa* allows the pronoun *él* to be interpreted as referring either to the subject antecedent or to the object antecedent. A subject-oriented interpretation corresponds to high attachment, while an object-oriented interpretation corresponds to low attachment.

A total of 16 experimental items were constructed, each with four variations. These variations were included to control for the possibility that specific proper names or character pairings could influence participants' interpretations. For Experiments 1 and 3, all experimental sentences were presented in Spanish; these materials are listed in Appendix A.2. For Experiment 2, structurally equivalent Danish sentences were used, preserving the same visual configuration and referential structure as far as possible; these materials are listed in Appendix A.3. The Danish materials were validated in consultation with Susi Olsen, a native expert linguist, to ensure grammatical naturalness and structural comparability.

For the Spanish experiments (Experiments 1 and 3), auditory stimuli were recorded by a native male speaker of Spanish. For the Danish baseline (Experiment 2), structurally equivalent Danish auditory stimuli were recorded by a native speaker of Danish. Recordings were produced with controlled intonation and a consistent speaking rate across items, with an approximate duration of seven seconds per sentence. Audio files were normalized for volume to ensure a consistent and comfortable listening level across trials.

Items were carefully controlled to minimize lexical biases and unintended cues that could favor one interpretation over another. In particular, the experimental sentences were designed so that both antecedents were grammatically possible referents for the pronoun. Visual scenes were constructed to include both potential antecedents and a scenery/background element (Appendix A.4), allowing fixations to be assigned to referential and non-referential regions.

In addition to the 16 experimental items, 24 filler sentences were included; these are listed together with the experimental materials in Appendices A.2 and A.3. Fillers were used to reduce predictability and discourage participants from developing task-specific strategies. Three training items were included at the beginning of each experiment to familiarize participants with the task structure. Training trials were excluded from all analyses.

5.3 Procedure

Participants completed the experiment individually in a quiet testing environment. Stimuli were presented on a computer screen while eye movements were recorded using an SR Research EyeLink system.

Each trial began with a fixation cross at the center of the screen. The fixation cross remained visible until the participant's gaze was detected within the target area, ensuring a standardized starting position across trials. Following successful fixation, the visual stimulus and auditory sentence were presented simultaneously. No preview of the image was provided prior to sentence onset.

The simultaneous presentation of visual and auditory stimuli was chosen to ensure that referential interpretation unfolded alongside the linguistic input. This design reduces the likelihood that participants formed expectations based on the visual scene before hearing the sentence, and it supports the interpretation of gaze behavior as reflecting online processing.

Participants viewed the scene while listening to the sentence, with eye movements recorded continuously throughout the trial. After each trial, a comprehension question was presented requiring participants to indicate which referent they interpreted the pronoun as referring to. Participants answered by clicking on the corresponding image region with the mouse. Including comprehension questions after all trials ensured sustained engagement and provided an explicit offline measure of interpretation.

No feedback was provided after comprehension responses. This was done to avoid encouraging participants to adjust their strategy during the experiment. The order of trials was pseudo-randomized, with the constraint that no more than two experimental trials appeared consecutively. Each session lasted approximately 10–15 minutes.

5.4 Eye-Tracking Apparatus and Recording Parameters

Eye movements were recorded monocularly from the right eye at a sampling rate of 1000 Hz using an SR Research EyeLink Duo system with a chinrest. Stimuli were presented on a 27-inch monitor with a resolution of 1920 $\times$ 1080 pixels. Participants were seated approximately 30 cm from the screen, with viewing distance and head position controlled using the chinrest. This setup reduced head movement and improved the reliability of gaze-position measurements.

A nine-point calibration procedure was conducted at the start of each session and repeated when necessary to ensure tracking accuracy. Calibration was used to ensure that recorded gaze coordinates could be reliably mapped onto the regions of interest in the visual display.

5.5 Regions of Interest and Gaze Mapping

Regions of interest (ROIs) were manually defined using screen-based coordinates. Three regions were specified for each display: the subject character, the object character, and the background/scenery region. Representative examples of the visual displays and ROI layout are provided in Appendix A.4.

ROIs were equal in size and spatially symmetrical across trials, ensuring that neither referent was visually advantaged by differences in region size or position. This control minimizes perceptual biases and helps ensure that gaze differences reflect linguistic processing rather than visual salience.

Fixations were assigned to ROIs based on their spatial coordinates, enabling the analysis of gaze allocation to each referent over time. ROI labels were interpreted according to the grammatical role of the corresponding character in the sentence, allowing fixation allocation to be compared in terms of subject- and object-related regions.

The background/scenery region was included as a non-antecedent region. This made it possible to distinguish fixations directed toward potential referents from fixations directed toward other visually present but non-referential elements of the scene.

5.6 Data Processing and Dependent Measures

Raw eye-tracking data were parsed using standard EyeLink algorithms. Fixations shorter than 80 ms were removed, as these were interpreted as reflecting rapid visual scanning rather than meaningful attentional selection. Fixations longer than 2000 ms were also excluded to reduce the influence of extreme outliers.

The primary dependent measure reported in the present analysis was the proportion of fixations directed to each ROI. Fixation counts were used to compute these proportions, allowing gaze allocation to be compared across subject, object, and scenery regions while normalizing for differences in overall fixation frequency.

Increased attention to subject-related regions was interpreted as evidence of high-attachment processing, while increased attention to object-related regions was interpreted as evidence of low-attachment processing. Fixations directed to the scenery region were treated as attention to a non-antecedent region and were used descriptively to characterize the overall distribution of gaze.

Comprehension responses were analyzed as an offline measure of interpretation, providing complementary evidence alongside the eye-tracking data. Whereas fixation proportions provide an online measure of attentional allocation during sentence processing, comprehension responses reflect the participant's explicit interpretation after the sentence had been processed.

5.7 Experimental Design Considerations

The study employed comparable materials and identical procedures across groups to ensure comparability. Simultaneous presentation of auditory and visual stimuli ensured that referential interpretation unfolded incrementally, while fixation-contingent trial onset standardized gaze starting position across trials.

Filler items and pseudo-randomization reduced strategic processing, and the inclusion of comprehension questions after each trial ensured participant engagement. Together, these design choices provide a controlled and reliable framework for investigating attachment ambiguity in real time.

The design also allows for comparison between online and offline measures. Fixation behavior provides information about the distribution of attention during sentence processing, while comprehension responses provide an explicit indication of final interpretation. This combination is particularly useful for investigating attachment ambiguity, as online gaze behavior and offline interpretation may reveal complementary aspects of the resolution process.

6 Results

This chapter presents the results of three eye-tracking experiments investigating attachment ambiguity resolution in native Spanish speakers, native Danish speakers, and Danish learners of Spanish. The results are organized by experiment, with each section reporting both fixation-based measures of visual attention and comprehension question responses. Fixation data are interpreted as indices of online processing, while comprehension responses provide complementary offline evidence of final interpretation.

The main focus of the analysis is the distribution of fixations across regions of interest corresponding to the subject referent, the object referent, and the scenery/background region. In the experimental design, increased fixation allocation to the subject-related region is interpreted as evidence for high-attachment processing, whereas increased fixation allocation to the object-related region is interpreted as evidence for low-attachment processing. Fixations to the scenery region are treated descriptively as attention to non-antecedent visual information.

6.1 Data Preprocessing

Prior to analysis, eye-tracking data were preprocessed to remove unreliable fixations. Fixations falling outside the display boundaries were excluded. In addition, fixations shorter than 80 ms were removed, as such brief fixations are unlikely to reflect meaningful attentional allocation and are typically associated with rapid visual scanning. Fixations longer than 2000 ms were also excluded to reduce the influence of extreme outliers.

After preprocessing, the remaining dataset was used for all analyses. Duration-based preprocessing was applied consistently in order to ensure that the fixation data reflected meaningful attentional allocation rather than rapid scanning or extreme fixation artifacts. The same preprocessing principles were applied across experiments in order to preserve comparability between participant groups.

6.2 Statistical Analysis

The primary goal of the statistical analysis was to determine whether participants exhibited a reliable preference for one referent over another during attachment ambiguity resolution. To this end, fixation proportions directed to subject- and object-related regions were compared within participants.

A paired-samples t-test was used to assess differences in fixation proportions between regions of interest. This test was selected because the design involves within-subject comparisons, where each participant contributes fixation data for both subject and object regions across trials. The paired structure of the data makes the t-test an appropriate method for evaluating mean differences between conditions.

The paired-samples t-tests were computed from participant-level fixation proportions available during the original analysis. However, the exported datasets available for the present descriptive summaries were not equally granular across all participant groups, which limits the possibility of reconstructing detailed participant-by-participant fixation-preference tables for the full dataset.

Fixation proportions were used rather than raw fixation counts in order to normalize for individual differences in overall eye-movement behavior. This normalization ensures that comparisons reflect relative attentional allocation rather than absolute differences in fixation frequency.

Given the relatively small sample sizes (Spanish: $n = 10$, Danish: $n = 10$, L2: $n = 6$), the statistical analysis is interpreted with caution. While the paired t-test provides a useful measure of group-level tendencies, it may have limited power to detect smaller effects, particularly in the L2 group. For this reason, statistical results are complemented by descriptive summaries and aggregate fixation patterns, which provide additional context for interpreting group-level tendencies.

No correction for multiple comparisons was applied, as the three comparisons were planned a priori and corresponded directly to the three participant groups specified in the research questions. All tests were two-tailed, and the significance level was set at $\alpha = 0.05$.

It is important to note that the present analysis focuses on aggregated fixation proportions and does not model time-course effects or trial-level variability. While more complex approaches such as mixed-effects modeling could provide a more fine-grained analysis, these methods typically require larger datasets and consistently exported participant- and item-level fixation data. Given the scope and structure of the available dataset, the chosen analytical approach provides a transparent and interpretable first-pass analysis of attachment preferences.

6.3 Descriptive Overview

Before presenting the results for each experiment separately, fixation counts and proportions were summarized across participant groups. This descriptive overview provides an initial account of how visual attention was distributed across the subject, object, and scenery regions.

Group	Subject region	Object region	Scenery region
Spanish baseline	1510 (50.8%)	688 (23.1%)	776 (26.1%)
Danish baseline	961 (32.3%)	1123 (37.8%)	890 (29.9%)
Danish L2 Spanish	1250 (42.0%)	1163 (39.1%)	561 (18.9%)

Table 6.1.: Total fixation counts and aggregate fixation proportions by region across participant groups.

As shown in Table 6.1, the Spanish baseline displayed the clearest asymmetry in fixation allocation, with approximately half of all fixations directed to the subject region. The Danish baseline showed a more balanced distribution between subject and object regions, with a slight numerical advantage for the object region. The Danish L2 Spanish group showed an intermediate pattern, with similar proportions of fixations to subject and object regions and fewer fixations to the scenery region.

This descriptive pattern provides an initial indication that the three groups differed in how visual attention was allocated across potential antecedents. However, these aggregate proportions should be interpreted together with the inferential analyses reported below, since aggregate counts alone do not indicate whether differences between regions were reliable at the group level.

Group	Subject response	Object response
Spanish baseline	67%	33%
Danish baseline	48%	52%
Danish L2 Spanish	54%	46%

Table 6.2.: Aggregate comprehension question responses by participant group.

Table 6.2 summarizes the comprehension question responses across groups. The Spanish baseline showed a clear preference for subject-oriented interpretations, consistent with the fixation-based evidence for high attachment. The Danish baseline showed a near-balanced distribution of responses, with a slight numerical preference for object-oriented interpretations. The Danish L2 Spanish group showed a modest subject-oriented tendency, although the response distribution remained more balanced than in the Spanish baseline.

The comprehension responses are important because they provide an offline measure of participants' final interpretation of the ambiguous pronoun. In this sense, they complement the eye-tracking data: whereas fixation proportions reflect online attentional allocation during sentence processing, comprehension responses indicate the interpretation selected after the trial. Agreement between these two measures strengthens the interpretation of an attachment preference, while divergence or weak patterns suggest greater uncertainty or instability.

6.4 Participant-Level Considerations

Because the available exported fixation data were not equally granular across all participant groups, participant-level fixation preference analyses were not conducted as part of the main results. The primary fixation analyses therefore focus on aggregate region-level fixation proportions, which were available consistently for the Spanish baseline, Danish baseline, and Danish L2 Spanish group.

This decision ensures that comparisons across groups are based on equivalent data structures. While participant-level analyses would provide additional insight into individual variability, especially in the L2 group, such analyses could not be performed consistently across the full dataset. Consequently, individual differences are treated as a limitation of the present analysis and are returned to in the Discussion.

6.5 Experiment 1: Spanish Baseline

Experiment 1 examined attachment ambiguity resolution in native speakers of Spanish. This experiment served as the target-language baseline against which the Danish L2 Spanish group could be compared. Based on previous findings on Spanish attachment preferences, Hypothesis H1 predicted that native Spanish speakers would show a high-attachment preference, operationalized as increased fixation allocation to the subject-related region.

6.5.1 Fixation Distribution

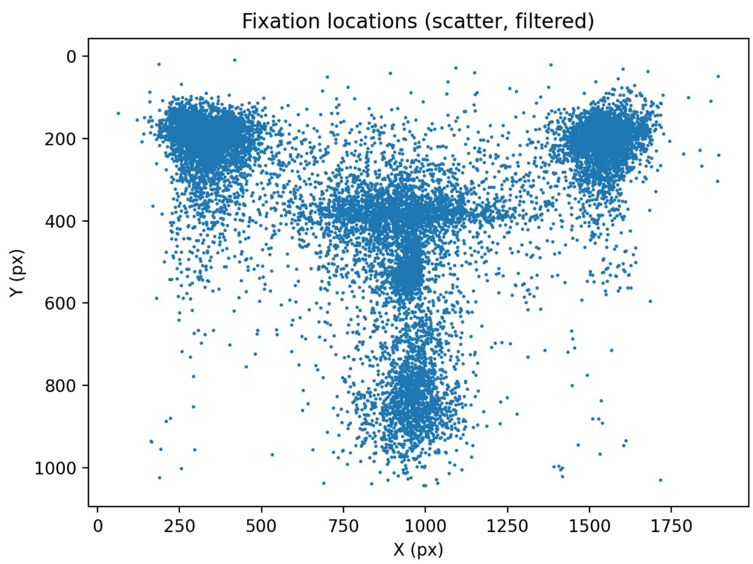


Figure 6.1.: Scatter distribution of fixation locations in the Spanish baseline.

Figure 6.1 shows the spatial distribution of fixation locations aggregated across participants and trials. Fixations are concentrated in three primary regions corresponding to the subject referent, the object referent, and the bottom scenery. Minimal fixation activity is observed outside these regions, indicating that participants consistently attended to task-relevant areas.

The clustering of fixations around the predefined regions of interest suggests that the visual display was processed in a task-relevant manner. This is important because the interpretation of fixation allocation relies on the assumption that participants attended primarily to the potential antecedents and relevant visual context, rather than to unrelated parts of the display.

6.5.2 Fixation Allocation

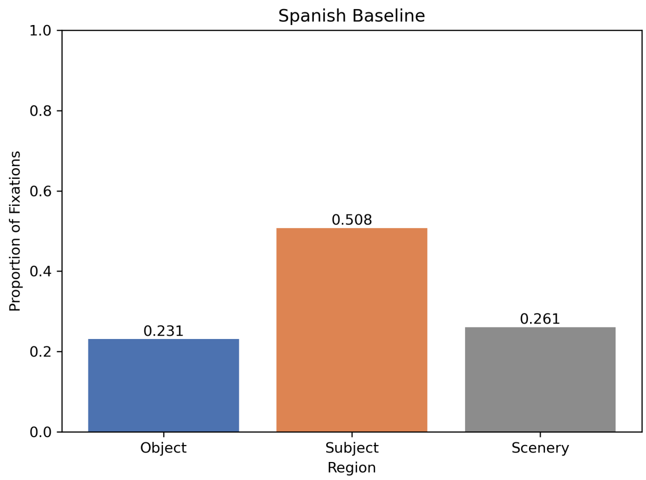


Figure 6.2.: Aggregate proportion of fixations by region in the Spanish baseline.

Figure 6.2 shows the proportion of fixations directed to each region of interest. Native Spanish speakers allocated 50.8% of fixations to the subject-related region, 23.1% to the object-related region, and 26.1% to the bottom scenery.

This distribution reveals a clear and substantial attentional bias toward the subject referent. Since the subject region corresponds to the high-attachment interpretation in the experimental design, increased fixation allocation to this region provides online evidence that Spanish participants preferentially considered the subject antecedent during ambiguity resolution.

A paired-samples t-test confirmed that the difference in fixation proportions between subject and object regions was statistically significant, $t(9) = 3.67$, $p = 0.005$. This result provides clear evidence for a high-attachment preference in the Spanish baseline under the present experimental conditions. Hypothesis H1 is therefore supported.

6.5.3 Comprehension Question Responses

Comprehension question responses showed a clear preference for subject-oriented interpretations. The majority of responses selected the subject referent, aligning with the fixation-based results. This convergence between online and offline measures provides converging evidence for a high-attachment preference in this group.

The agreement between fixation allocation and comprehension responses suggests that the subject-oriented pattern was not limited to transient gaze behavior, but was also reflected in participants' final interpretation of the ambiguous pronoun. The Spanish baseline therefore provides a clear target-language reference point for the L2 comparison.

6.6 Experiment 2: Danish Baseline

Experiment 2 examined attachment ambiguity resolution in native speakers of Danish. This experiment was included to establish an L1 baseline for the Danish learners of Spanish. Hypothesis H2 predicted that Danish speakers would show a low-attachment preference, operationalized as increased fixation allocation to the object-related region.

6.6.1 Fixation Distribution

Figure 6.3 presents the spatial distribution of fixations for Danish native speakers. As in the Spanish baseline, fixations are concentrated in the subject, object, and scenery regions, indicating that participants attended to the relevant elements of the visual display.

The scatter distribution shows that Danish participants engaged with the relevant regions of the display, but the spatial distribution does not by itself indicate a clear attachment preference. For this reason, fixation proportions across the subject and object regions were examined more directly.

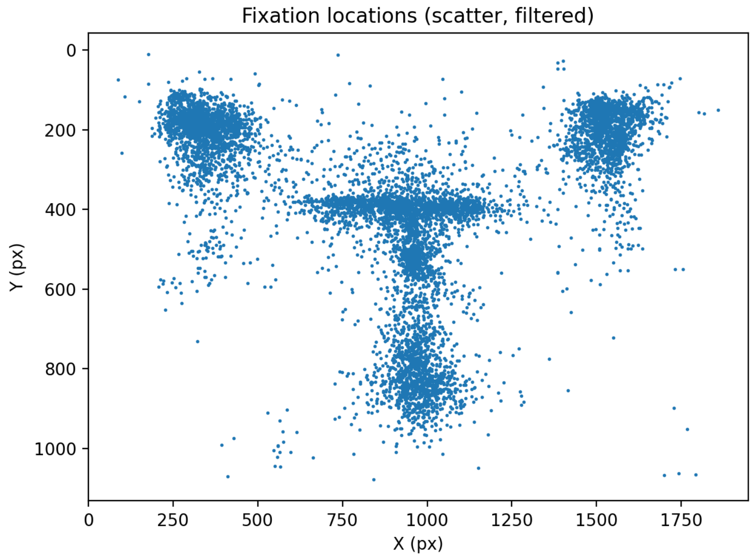


Figure 6.3.: Scatter distribution of fixation locations in the Danish baseline.

6.6.2 Fixation Allocation

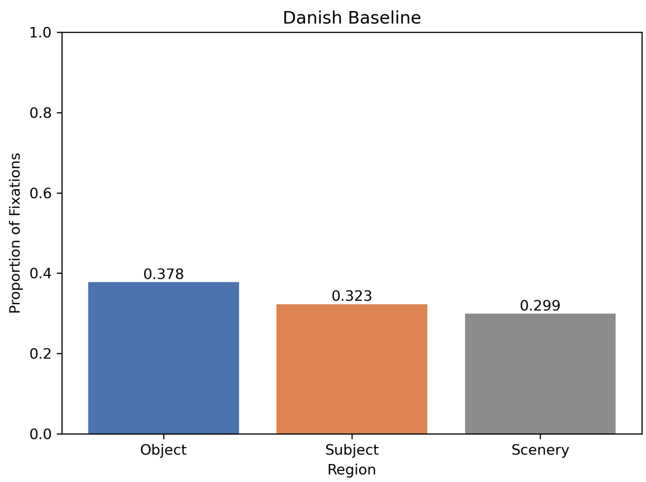


Figure 6.4.: Aggregate proportion of fixations by region in the Danish baseline.

Figure 6.4 shows the proportion of fixations directed to each region of interest. Native Danish speakers allocated 32.3% of fixations to the subject-related region, 37.8% to the object-related region, and 29.9% to the bottom scenery.

This distribution indicates a relatively balanced allocation of attention between subject and object referents, with only a small numerical advantage for the object region. Although this numerical pattern is in the direction of a low-attachment preference, the difference between subject and object regions is modest and does not suggest a strong or consistent attachment preference at the group level.

A paired-samples t-test did not reveal a significant difference between fixation proportions for subject and object regions, $t(9) = 0.39$, $p = .705$. This result does not support Hypothesis H2. Rather than showing a reliable low-attachment preference, Danish participants showed a relatively balanced fixation pattern under the present experimental conditions.

The absence of a significant difference should not be interpreted as evidence that Danish speakers have no attachment preferences in general. Instead, it indicates that the present materials and task did not elicit a reliable group-level preference in the Danish baseline.

6.6.3 Comprehension Question Responses

Comprehension responses mirrored the fixation data. Participants did not consistently favor either subject- or object-oriented interpretations, indicating that no stable attachment preference emerged at the level of explicit judgments.

This alignment between the eye-tracking and comprehension data strengthens the interpretation that Danish attachment preferences were weak or unstable in this task. Both online and offline measures point toward a relatively balanced pattern rather than a clear low-attachment strategy.

6.7 Experiment 3: Danish Learners of Spanish

Experiment 3 examined attachment ambiguity resolution in Danish learners of Spanish. This group was compared with both the Spanish baseline and the Danish baseline in order to assess whether learner behavior aligned more closely with the target language or with the first language. Hypothesis H3a predicted convergence toward the Spanish high-attachment pattern, whereas Hypothesis H3b allowed for L1 transfer or unstable processing patterns.

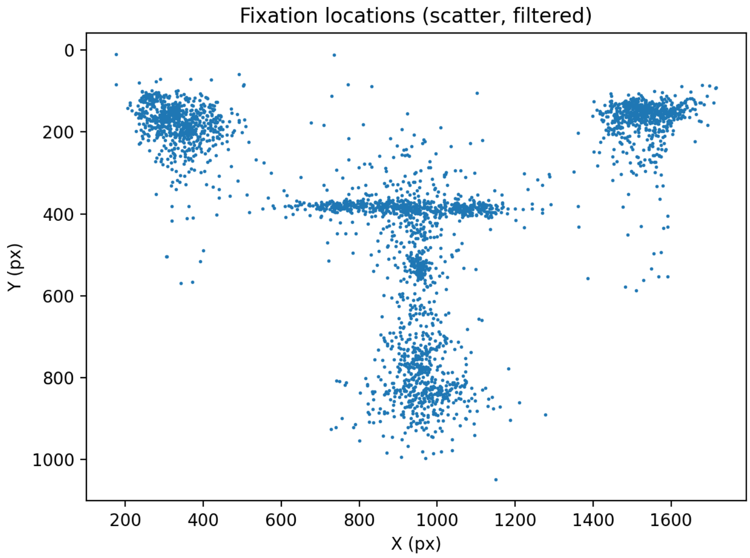


Figure 6.5.: Scatter distribution of fixation locations in the Danish L2 Spanish group.

6.7.1 Fixation Distribution

Figure 6.5 shows the spatial distribution of fixations for Danish learners of Spanish. As in the baseline experiments, fixations cluster around the subject, object, and scenery regions. However, the distribution appears more dispersed, suggesting greater variability in attentional allocation at the aggregate level.

As with the two baseline groups, the scatter plot indicates that learners attended primarily to task-relevant regions. However, the spatial distribution alone does not provide sufficient evidence for a clear attachment preference, and fixation proportions were therefore examined directly.

6.7.2 Fixation Allocation

Figure 6.6 shows the proportion of fixations directed to each region of interest. Danish learners of Spanish allocated 42.0% of fixations to the subject-related region, 39.1% to the object-related region, and 18.9% to the bottom scenery.

This distribution indicates a slight tendency toward subject-oriented processing, although the difference between subject and object regions is relatively small. Compared to the Spanish baseline, the subject preference is markedly weaker. Compared to the Danish baseline, however, the L2 group shows a more subject-

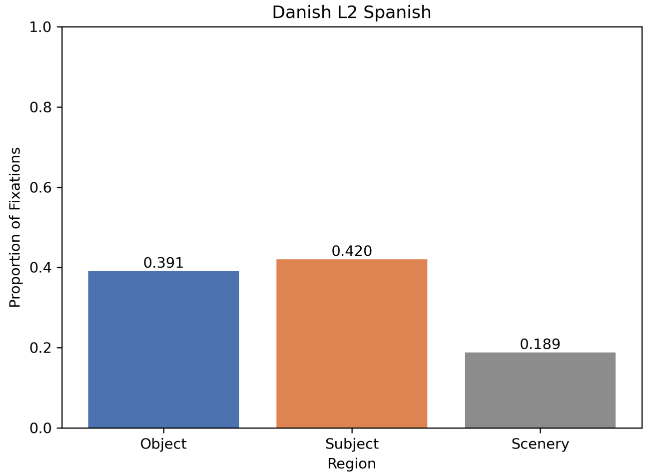


Figure 6.6.: Aggregate proportion of fixations by region in the Danish L2 Spanish group.

oriented pattern. In descriptive terms, the L2 group therefore appears to fall between the two baseline groups.

A paired-samples t-test did not reveal a significant difference between subject and object fixation proportions, $t(5) = 1.02$, $p = .334$. This lack of statistical significance reflects the relatively small difference between subject and object fixation proportions, together with the limited sample size in the L2 group.

The L2 pattern is therefore best interpreted descriptively. The learners showed a slight numerical movement toward the Spanish subject-oriented pattern, but this tendency was not strong enough to support a reliable group-level high-attachment preference. Consequently, Hypothesis H3a is not clearly supported. At the same time, the L2 pattern does not provide strong evidence for straightforward L1 transfer either, since the learners did not pattern identically with the Danish baseline.

6.7.3 Comprehension Question Responses

Comprehension responses in the L2 group were similarly mixed at the aggregate level, with a modest numerical preference for subject-oriented interpretations but no clear categorical dominance.

This pattern is broadly consistent with the fixation data: both measures suggest a weak subject-oriented tendency, but neither provides evidence for a strong or stable high-attachment preference. The L2 results therefore suggest partial movement toward the Spanish baseline while remaining weaker and less clearly defined than the native Spanish pattern.

6.8 Additional Data Quality and Exploratory Checks

6.8.1 Danish Baseline Fixation Characteristics

For the Danish baseline, fixation events were extracted from the converted EyeLink ASCII files. These fixation-level data were used to inspect general fixation characteristics, including fixation duration, fixation count, and duration-based exclusions. These analyses were treated as data-quality checks rather than as the basis for additional participant-level attachment analyses, since equivalent fixation-level exports were not available for all participant groups.

Across the ten Danish participants, 13,895 raw fixation events were identified. Of these, 12,416 occurred after image display onset and were therefore relevant for the visual-world analysis. Fixations occurring before image display were excluded from the fixation-based analyses.

Duration-based preprocessing was then applied to the post-image-display fixations. In total, 379 fixations shorter than 80 ms and 52 fixations longer than 2000 ms were excluded, corresponding to 431 fixations, or 3.47% of the post-image-display fixation data. The final cleaned Danish baseline dataset therefore contained 11,985 valid fixations.

Before duration-based preprocessing, the mean fixation duration was 338.49 ms ($SD = 302.49$), with durations ranging from 3 ms to 5576 ms. After preprocessing, the mean fixation duration was 336.65 ms ($SD = 247.93$), with durations ranging from 80 ms to 2000 ms. This indicates that preprocessing removed extreme values while preserving the majority of fixation data.

Fixation counts were also summarized by trial and participant. After preprocessing, the Danish baseline contained an average of 27.87 fixations per trial

Dataset	N fixations	Mean	SD	Min	Max
Before duration cleaning	12416	338.49	302.49	3	5576
After duration cleaning	11985	336.65	247.93	80	2000

Table 6.3.: Fixation (ms) duration characteristics for the Danish baseline before and after duration-based preprocessing.

Exclusion criterion	Excluded fixations	Percentage of post-image fixations
Short fixations (< 80 ms)	379	3.05%
Long fixations (> 2000 ms)	52	0.42%
Total duration-based exclusions	431	3.47%

Table 6.4.: Fixations excluded from the Danish baseline by duration-based preprocessing criteria.

($SD = 6.65$). At the participant level, the average number of valid fixations was 1198.50 ($SD = 181.02$), with individual fixation counts ranging from 807 to 1495.

Measure	Mean	SD	Min	Median	Max
Fixations per trial	27.87	6.65	4	28.00	55
Fixations per participant	1198.50	181.02	807	1195.50	1495

Table 6.5.: Fixation count summary for the Danish baseline after preprocessing.

These checks indicate that the Danish baseline contained a substantial amount of usable fixation data after preprocessing. Although participant-level attachment preferences were not analyzed from these fixation-level exports, the fixation counts suggest that the aggregate Danish results were not driven by an absence of usable gaze data.

6.8.2 Trial and Character Effects

Trial-level time-course effects and character-specific effects were not tested statistically in the present analysis. This decision was made because the available fixation data were not equally granular across all participant groups, and because the sample sizes were limited. Consequently, no direct conclusions are drawn regarding trial-order effects or character-specific effects. However, several design choices were implemented to reduce the likelihood that such factors drove the main findings.

First, regions of interest were manually defined to be equal in size and spatially symmetrical across trials. Second, the same visual structure was used across experiments, ensuring comparability across participant groups. Third, filler items and pseudo-randomization were used to reduce strategic adaptation during the experiment. Finally, comprehension questions were included after each trial to maintain participant engagement.

Consequently, the results should be interpreted as group-level attachment patterns under the present experimental conditions rather than as a full model of trial-by-trial or character-specific variation. Future analyses should export fixation-level ROI data consistently across all participant groups, allowing participant-, item-, trial-order-, and character-level variability to be modeled directly using mixed-effects approaches.

6.9 Cross-Group Comparison

A comparison of fixation proportions across the three participant groups reveals distinct descriptive patterns in attachment-related gaze behavior.

Native Spanish speakers showed a strong subject-oriented bias, allocating 50.8% of fixations to the subject region compared to 23.1% for the object region. In contrast, native Danish speakers displayed a more balanced pattern, with 32.3% of fixations directed to the subject region and 37.8% to the object region.

Danish learners of Spanish exhibited an intermediate pattern. Fixations were distributed with 42.0% directed toward the subject region and 39.1% toward the object region. This distribution reflects a slight tendency toward subject-oriented processing, but the difference between regions is substantially smaller than in the Spanish baseline.

Taken together, these results suggest a gradient pattern across groups. Spanish speakers show a clear high-attachment preference, Danish speakers show no reliable preference, and L2 learners fall between these two baselines. This intermediate pattern is consistent with partial convergence toward the target language, although participant-level variability cannot be evaluated directly from the available aggregate fixation data.

6.10 Summary of Results

The results reveal clear differences across participant groups. Native Spanish speakers exhibited a clear high-attachment preference, reflected consistently in both fixation behavior and comprehension responses. In contrast, native Danish speakers showed no reliable attachment preference, with fixation allocation distributed relatively evenly across referents.

Danish learners of Spanish displayed no consistent group-level attachment preference. Their behavior does not fully align with either baseline group, suggesting partial convergence on L2 processing strategies while leaving open the possibility of L1 influence and individual variability. The additional data-quality checks for the Danish baseline further indicate that the absence of a robust Danish attachment preference cannot be attributed simply to a lack of usable fixation data.

7 Discussion

This thesis examined attachment ambiguity resolution in native speakers of Spanish, native speakers of Danish, and Danish learners of Spanish using eye-tracking measures within a visual-world paradigm. The main aim was to assess whether cross-linguistic differences in attachment preferences are reflected in online attentional patterns and to examine how such preferences manifest in second language processing.

The study was organized around three research questions. First, it asked whether native speakers of Spanish show a high-attachment preference when processing ambiguous pronoun reference in the present materials. Second, it examined whether native speakers of Danish show a comparable or different attachment preference under equivalent experimental conditions. Third, it investigated whether Danish learners of Spanish resolve Spanish attachment ambiguities in a way that reflects convergence toward the Spanish baseline, transfer from Danish, or a more variable pattern.

Overall, the results suggest that attachment preferences are not uniform across the three groups. Native Spanish speakers showed a clear subject-oriented pattern, consistent with high attachment. Native Danish speakers did not show a reliable group-level attachment preference, instead displaying a relatively balanced distribution of fixations and comprehension responses. Danish learners of Spanish showed an intermediate aggregate pattern, with a weak numerical tendency toward subject-oriented processing, but without a statistically reliable group-level preference.

7.1 Overview of Findings

Three main findings emerged from the analyses. First, the Spanish baseline provided clear evidence for a high-attachment preference. Spanish speakers allocated a substantially larger proportion of fixations to the subject-related region than to the object-related region, and this pattern was also reflected in comprehension responses. This supports Hypothesis H1, which predicted increased attention to subject-related regions in the Spanish baseline.

Second, the Danish baseline did not provide evidence for a reliable low-attachment preference. Although fixation allocation showed a small numerical advantage for the object-related region, this difference was not statistically significant, and comprehension responses were similarly balanced. Hypothesis H2 was therefore not supported. Rather than patterning clearly with the expected low-attachment strategy, Danish speakers showed weak or unstable attachment behavior under the present experimental conditions.

Third, Danish learners of Spanish did not show full convergence on the Spanish high-attachment pattern, nor did they clearly replicate the Danish baseline. Their fixation pattern was descriptively intermediate: subject-related fixations were numerically higher than object-related fixations, but the difference was small and not statistically reliable. This means that Hypothesis H3a, which predicted convergence toward Spanish high attachment, was not clearly supported. At the same time, the learner pattern also does not provide strong evidence for direct L1 transfer, since the L2 group did not simply reproduce the Danish baseline pattern. The results are therefore most consistent with a cautious interpretation in which L2 attachment resolution is characterized by partial target-language alignment, weak group-level effects, and unresolved variability.

7.2 Spanish Baseline and High-Attachment Processing

The Spanish baseline produced the clearest result in the study. Native Spanish speakers directed 50.8% of fixations to the subject-related region and 23.1% to the object-related region. This difference was statistically significant and was accompanied by a clear subject preference in comprehension responses. Together, these results indicate that the Spanish participants preferentially interpreted the ambiguous pronoun as referring to the subject antecedent.

This finding is consistent with previous research reporting high-attachment preferences in Spanish and other Romance languages. Importantly, the present results suggest that this preference is not limited to explicit, offline interpretation. The eye-tracking data show that the subject-related antecedent attracted more attention during the task itself, indicating that the high-attachment preference was reflected in online attentional allocation. In this sense, the visual-world paradigm provides evidence that Spanish attachment preferences can shape real-time processing.

The convergence between fixation behavior and comprehension responses is particularly important. If Spanish participants had shown a subject preference only in comprehension responses, the result could have been interpreted as a post-processing decision or a response strategy. Conversely, if the fixation data had shown a subject bias without corresponding comprehension responses, the interpretation of the eye-tracking pattern would have been less clear. Instead, both measures point in the same direction. This strengthens the conclusion that the Spanish baseline reflects a stable high-attachment preference in the present materials.

The Spanish result also provides an important reference point for the L2 comparison. Since Danish learners of Spanish were tested in Spanish, the Spanish baseline represents the relevant target-language pattern. The fact that the native Spanish group showed such a clear subject-oriented preference means that the experimental materials were capable of eliciting the expected high-attachment pattern in the target language. This is methodologically important because it reduces the possibility that the absence of a clear L2 effect was simply due to the materials being unable to reveal attachment preferences.

7.3 Danish Baseline and the Absence of a Reliable Attachment Preference

The Danish baseline produced a different pattern. Native Danish speakers allocated 32.3% of fixations to the subject-related region and 37.8% to the object-related region. Although this distribution showed a slight numerical advantage for the object region, the difference was small and not statistically significant. Comprehension responses were also close to balanced, with 48% subject responses and 52% object responses.

This result does not support a strong low-attachment preference in Danish under the present experimental conditions. This is theoretically relevant because the Danish baseline was included partly to determine whether Danish would pattern like other Germanic languages that have often been associated with low-attachment preferences. The present data do not provide clear evidence for such a strategy. Instead, Danish attachment behavior appears weak, balanced, or task-sensitive in this paradigm.

There are several possible interpretations of this finding. One possibility is that Danish does not have a strong attachment preference for this type of ambiguous pronoun construction. If so, Danish speakers may rely less on a stable syntactic attachment heuristic and more on other sources of information, such as discourse context, lexical plausibility, or task-specific cues. Another possibility is that Danish speakers do have attachment preferences, but these preferences were not strongly elicited by the present visual-world task.

The latter possibility is important because task specificity may play a role in the results. In the visual-world paradigm used here, both potential antecedents were visually available throughout the trial. This may reduce the need to commit strongly to one interpretation, since participants can maintain attention to multiple possible referents. In a purely linguistic task, such as reading without visual support or forced interpretation judgments, participants might show stronger attachment preferences. Therefore, the absence of a reliable Danish preference should not be interpreted as evidence that Danish speakers never display attachment biases. Rather, it indicates that no robust group-level preference emerged in the present task.

The Danish result is also important for interpreting the L2 findings. If Danish had shown a strong low-attachment preference, then L2 object-oriented behavior could have been interpreted more straightforwardly as evidence of L1 transfer. However, because the Danish baseline itself was relatively balanced, the interpretation of L1 influence becomes more complex. The L2 group cannot be evaluated against a clearly low-attaching Danish baseline; instead, Danish appears to provide a weak or unstable L1 reference point.

7.4 Attachment Resolution in Danish Learners of Spanish

The Danish L2 Spanish group showed an intermediate aggregate pattern. Learners allocated 42.0% of fixations to the subject-related region and 39.1% to the object-related region. Comprehension responses showed a similar weak subject-oriented tendency, with 54% subject responses and 46% object responses. However, the difference between subject and object fixation proportions was not statistically significant.

This pattern does not support a strong version of L2 convergence on the Spanish baseline. If Danish learners of Spanish had fully adopted the Spanish high-attachment preference, one would expect their fixation behavior to more closely resemble the native Spanish pattern, where subject-directed fixations were clearly dominant. Instead, the L2 group showed only a small numerical subject advantage. The learners therefore did not display a robust native-like high-attachment preference at the group level.

At the same time, the L2 results do not support a simple account based on direct transfer from Danish. The Danish baseline showed a slight numerical object advantage and no reliable preference, whereas the L2 group showed a slight numerical subject advantage. This means that the learners did not simply reproduce the Danish pattern. The L2 result is therefore best interpreted as lying between the two baselines: more subject-oriented than the Danish baseline, but substantially weaker than the Spanish baseline.

This intermediate pattern is theoretically interesting because it suggests that L2 attachment processing may not be adequately described in terms of either full convergence or direct transfer alone. Instead, L2 processing may involve a combination of developing sensitivity to target-language patterns, residual influence from the L1, and general processing constraints associated with real-time comprehension in a non-native language. The present data are compatible with such a mixed account, but they do not allow strong conclusions about the relative contribution of each factor.

The small L2 sample size is especially important here. With only six learners, the analysis has limited statistical power to detect subtle group-level effects. The absence of a significant subject-object difference therefore does not necessarily mean that learners have no emerging preference. Rather, it means that the present dataset does not provide sufficient evidence for a reliable group-level attachment bias. The descriptive subject tendency may reflect partial movement toward the Spanish pattern, but this interpretation must remain cautious.

The L2 findings also highlight the importance of individual differences. Factors such as proficiency, age of acquisition, amount of exposure, and frequency of Spanish use are likely to influence whether learners adopt target-language processing strategies. However, because participant-level fixation preferences could not be reconstructed consistently across all groups, the present study cannot directly evaluate such individual differences. This is an important limitation, particularly for the L2 group, where individual variability is theoretically expected to play a substantial role.

7.5 Online and Offline Measures

A central motivation for using eye tracking was to compare online attentional allocation with offline interpretation. The results show that these two measures were most clearly aligned in the Spanish baseline. Spanish participants showed both increased fixation allocation to the subject region and a higher proportion of subject responses in comprehension questions. This convergence strengthens the interpretation that the Spanish baseline reflects a stable high-attachment preference.

In the Danish baseline, online and offline measures were also broadly aligned, but in a different way. Both fixation allocation and comprehension responses were relatively balanced. This suggests that the absence of a reliable Danish attachment preference was not merely an artifact of one measure. Instead, both gaze behavior and explicit responses point toward weak or unstable attachment resolution under the present conditions.

For the L2 group, the relationship between online and offline measures was again broadly consistent. Both measures showed a modest numerical subject tendency, but neither indicated a strong or reliable group-level preference. This parallel between fixation proportions and comprehension responses suggests that the L2 pattern was not limited to momentary gaze behavior, but it also did not result in a strong final interpretive preference.

These findings illustrate why combining online and offline measures is useful. Eye-tracking data provide a graded measure of attentional allocation during sentence processing, while comprehension questions provide a categorical measure of final interpretation. In the present study, the combination of the two measures allowed for a more nuanced interpretation than either measure alone would have provided.

7.6 Methodological Considerations

The visual-world paradigm was well suited to the goals of the study because it allowed referential processing to be examined as participants listened to ambiguous sentences. By presenting both potential antecedents visually, the paradigm made it possible to measure how attention was distributed between competing referents during the task.

However, the visual-world paradigm also shapes the interpretation of the results. Because both antecedents were continuously visible, participants may have been able to maintain multiple candidate interpretations rather than committing immediately to one antecedent. This may be particularly relevant for the Danish baseline and the L2 group, where fixation allocation was relatively balanced. In these cases, the visual context may have encouraged distributed attention across both referents.

This does not undermine the usefulness of the paradigm, but it means that the results should be interpreted as attachment preferences under the present visual-world conditions. The task may not capture exactly the same processes as silent reading, auditory comprehension without images, or forced-choice interpretation tasks. Instead, it provides a controlled way of examining how linguistic ambiguity interacts with visual attention when multiple possible referents are simultaneously available.

Another methodological consideration concerns the interpretation of fixations. Increased fixation to a referent is treated here as evidence that the corresponding antecedent is being considered during interpretation. This is a standard assumption in visual-world studies, but fixation behavior is not a direct measure of syntactic representation. It reflects the allocation of visual attention, which may be influenced by linguistic processing, visual salience, task demands, and response preparation. For this reason, fixation results were interpreted together with comprehension responses rather than in isolation.

The additional data-quality checks for the Danish baseline further support the reliability of the Danish eye-tracking data. The converted EyeLink ASCII files showed that the Danish baseline contained a substantial number of usable fixations after preprocessing, and only a small proportion of post-image-display fixations were excluded due to duration thresholds. This suggests that the absence of a reliable Danish attachment preference cannot simply be attributed to insufficient usable gaze data. However, equivalent fixation-level exports were not available for all groups, so these checks were treated as data-quality evidence rather than as the basis for additional cross-group analyses.

7.7 Limitations

The present study is subject to several limitations. The most important limitation is sample size. The Spanish and Danish baselines each included ten participants, while the L2 group included six participants. These sample sizes are relatively small, especially for detecting subtle differences in fixation behavior. As a result, non-significant findings should be interpreted cautiously. This is particularly important for the L2 group, where the observed numerical subject tendency may reflect an emerging pattern that the present study was underpowered to detect reliably.

A second limitation concerns the granularity of the exported eye-tracking data. The main fixation analyses were conducted at the aggregate region level because equivalent participant-level fixation exports were not available consistently across all groups. This limits the extent to which participant-level variation could be examined. As a result, the study cannot determine whether the aggregate L2 pattern reflects a broadly weak group-level tendency or a mixture of different individual strategies.

A third limitation concerns item- and character-level effects. Trial-level time-course effects and character-specific effects were not statistically modeled in the present analysis. Although the stimuli were designed with symmetrical ROIs, comparable visual layouts, filler items, and pseudo-randomization, the present analysis cannot fully rule out the possibility that particular trials or character pairings influenced fixation behavior. Future work should model participant- and item-level variability directly.

A fourth limitation concerns proficiency and exposure in the L2 group. Although the learners were characterized as intermediate to advanced speakers of Spanish, the present analysis did not model proficiency as a predictor of attachment behavior. Since L2 processing is likely to be shaped by proficiency, exposure, and frequency of use, this limits the conclusions that can be drawn about the development of native-like attachment preferences in Danish learners of Spanish.

Finally, the study focused on one specific type of ambiguity: pronoun-based attachment ambiguity involving subject and object antecedents. Attachment preferences may differ across constructions, discourse contexts, and experimental tasks. Therefore, the results should not be generalized to all forms of ambiguity resolution without further evidence.

7.8 Future Directions

Future research should expand the present design in several ways. First, larger participant samples are needed, especially for the L2 group. A larger learner sample would provide greater statistical power and would make it possible to examine individual differences in proficiency, exposure, and age of acquisition.

Second, future studies should export fixation-level ROI data consistently across all participant groups. This would allow participant-level and item-level analyses to be conducted directly. Mixed-effects models could then be used to account for variability across participants, items, trial order, and character identity. Such analyses would provide a more precise account of the factors influencing attachment resolution.

Third, future research could examine the time course of attachment resolution more directly. The present study analyzed aggregate fixation proportions, but attachment preferences may unfold dynamically over the course of a trial. For example, participants might initially attend to one antecedent before later shifting to another. Time-window analyses or growth-curve approaches could help determine whether attachment preferences emerge early, late, or only after the ambiguous pronoun has been processed.

Fourth, future studies could compare the visual-world paradigm with other experimental methods. If Danish speakers show weak attachment preferences in the visual-world paradigm but stronger preferences in reading or forced-choice tasks, this would suggest that task demands play an important role in attachment resolution. Such comparisons would help clarify whether the Danish pattern observed here reflects a genuine absence of a strong preference or a task-specific attenuation of attachment biases.

Finally, further work could examine other language pairings and other ambiguity types. Comparing Danish learners of Spanish with learners from languages that have clearer low-attachment preferences would help determine whether L1 transfer effects are stronger when the L1 baseline is itself more stable. Extending the paradigm to other structures would also help establish whether the patterns observed here are specific to pronoun-based attachment ambiguity or reflect broader principles of L2 sentence processing.

7.9 Conclusion of the Discussion

Taken together, the findings suggest that attachment preferences are both language-specific and task-sensitive. The Spanish baseline showed a clear high-attachment preference, supporting previous findings on Spanish and demonstrating that the present materials successfully elicited the expected target-language pattern. The Danish baseline, in contrast, did not show a reliable attachment preference, suggesting that Danish attachment resolution in this context may be weak, unstable, or sensitive to the visual-world task.

The Danish learners of Spanish showed an intermediate pattern that did not fully align with either baseline group. This suggests that L2 attachment resolution cannot be explained solely in terms of full convergence on the target language or direct transfer from the first language. Instead, the results point toward a more nuanced account in which L2 processing reflects partial target-language alignment, limited statistical power, task-specific processing demands, and unresolved individual variability.

8 Conclusion

This thesis examined the resolution of attachment ambiguities involving pronoun reference in native speakers of Spanish, native speakers of Danish, and Danish learners of Spanish. Using eye-tracking measures within a visual-world paradigm, the study investigated whether cross-linguistic differences in attachment preferences are reflected in online attentional patterns and how such preferences manifest in second language processing.

The results showed that native speakers of Spanish exhibited a clear high-attachment preference. This was reflected in both fixation-based measures and comprehension responses, indicating convergence between online attentional allocation and offline interpretation. This finding is in line with previous work on Spanish and supports the view that attachment preferences in this language are engaged during real-time sentence processing.

By contrast, native speakers of Danish did not display a reliable attachment preference under the present experimental conditions. Fixation allocation was relatively balanced across potential antecedents, and comprehension responses showed a similarly balanced pattern. This suggests that Danish attachment preferences in this task are weak, unstable, or sensitive to experimental conditions, rather than clearly reflecting a robust low-attachment strategy. This finding is particularly relevant given the limited amount of previous work on Danish attachment ambiguity resolution.

The Danish learners of Spanish showed an intermediate aggregate pattern. Their fixation allocation displayed a slight numerical tendency toward the subject-related region, but this tendency was considerably weaker than in the Spanish baseline and did not reach statistical significance. The learner group therefore did not show clear evidence of full convergence on the Spanish high-attachment preference. At the same time, their pattern did not straightforwardly replicate the Danish baseline either. These results suggest that L2 attachment resolution cannot be explained solely in terms of either target-language convergence or direct L1 transfer.

Overall, the findings indicate that attachment preferences vary across languages and that second language attachment resolution may involve partial convergence, weak transfer effects, task-specific processing demands, and unresolved individual variability. The study also demonstrates the usefulness of combining eye-tracking measures with comprehension responses, as this makes it possible to compare online attentional patterns with offline interpretation choices.

At the same time, the conclusions must be interpreted in light of the study's limitations. The sample sizes were relatively small, especially in the L2 group, and the available fixation data did not allow participant-level fixation preferences or item-level effects to be modeled consistently across all groups. Future research should therefore extend the present design with larger participant samples, consistently exported fixation-level ROI data, and statistical models capable of accounting for participant- and item-level variability.

Despite these limitations, the thesis contributes new empirical evidence on attachment ambiguity resolution in Danish and Danish L2 Spanish, two areas that remain underrepresented in the existing literature. By comparing Spanish, Danish, and Danish learners of Spanish within the same visual-world paradigm, the study provides a basis for further investigation of how language-specific parsing preferences interact with second language processing in real time.

9 Perspectives

Several directions for future research follow from the present study. First, larger participant samples, particularly in the second language group, would allow for more reliable statistical inference and a more detailed assessment of individual differences in attachment processing. Increased sample sizes would also make it possible to examine the role of factors such as proficiency, exposure, and age of acquisition.

Second, extending the experimental paradigm to include additional syntactic constructions and types of attachment ambiguity would help determine the extent to which the observed patterns generalize beyond the structures examined here. Attachment preferences may vary across constructions, and a broader range of stimuli would provide a more comprehensive picture of attachment strategies across languages.

Third, future work could compare visual-world paradigms with tasks that do not involve visual context, such as self-paced reading or acceptability judgment tasks. Such comparisons would help clarify the extent to which visual information influences attachment resolution and whether certain attachment preferences are attenuated or enhanced by the availability of visual cues.

Finally, longitudinal approaches could be used to investigate how attachment resolution strategies develop over time in second language acquisition. Tracking changes in fixation behavior as learners gain experience with the target language would contribute to a more precise understanding of the dynamics of convergence and variability in second language sentence processing.

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Appendices

A

A.1 English Reference Sentences

The following table reports the English reference sentences associated with the experimental materials. These English sentences were not used as experimental stimuli in the reported experiments, but are included here for documentation and transparency.

Table A.1.: English reference sentences, conditions, and comprehension questions for the experimental, filler, and practice materials.

Type	Condition	Sentence	Comprehension question
experimental	d	Peter hugged Mrs. Jones by the tree. Immediately afterwards he picked a couple of green leaves.	Who picked the green leaves?
experimental	a	Mr. Smith helped Peter by the table. Immediately afterwards he ate a couple of nice steaks.	Who ate the steaks?
experimental	b	Mrs. Jones thanked Peter by the car. Immediately afterwards he held a couple of metal nuts.	Who held the nuts?
experimental	c	Peter brought Mr. Smith to the museum. Immediately afterwards he bought a couple of postcards.	Who bought the postcards?
experimental	d	Peter helped Mrs. Jones by the door. Immediately afterwards he bought a couple of oil cans.	Who bought the oil cans?
experimental	a	Mr. Smith helped Peter by the house. Immediately afterwards he bought a couple of soap bars.	Who bought the soap bars?

Type	Condition	Sentence	Comprehension question
experimental	b	Mrs. Jones hugged Peter by the stove. Immediately afterwards he ate a couple of sandwiches.	Who ate the sandwiches?
experimental	c	Peter scolded Mr. Smith by the fridge. Immediately afterwards he ate a couple of pancakes.	Who ate the pancakes?
experimental	d	Peter helped Mrs. Jones in the office. Immediately afterwards he sent a couple of documents.	Who sent the documents?
experimental	a	Mr. Smith found Peter by the path. Immediately afterwards he drank a couple of glasses of water.	Who drank the water?
experimental	b	Mrs. Jones advised Peter by the tent. Immediately afterwards he bought a couple of oranges.	Who bought the oranges?
experimental	c	Peter helped Mr. Smith by the shop. Immediately afterwards he chose a couple of rugby balls.	Who chose the rugby balls?
experimental	d	Peter scolded Mrs. Jones by the sink. Immediately afterwards he cleaned a couple of forks.	Who cleaned the forks?
experimental	a	Mr. Smith found Peter at the bus stop. Immediately afterwards he bought a couple of tickets.	Who bought the tickets?
experimental	b	Mrs. Jones thanked Peter by the factory. Immediately afterwards he fixed a couple of lightbulbs.	Who fixed the lightbulbs?
experimental	c	Peter hugged Mr. Smith by the garden. Immediately afterwards he picked a couple of flowers.	Who picked the flowers?
experimental	c	Peter hugged Mr. Smith by the tree. Immediately afterwards he picked a couple of green leaves.	Who picked the green leaves?
experimental	d	Peter helped Mrs. Jones by the table. Immediately afterwards he ate a couple of nice steaks.	Who ate the steaks?

Type	Condition	Sentence	Comprehension question
experimental	a	Mr. Smith thanked Peter by the car. Immediately afterwards he held a couple of metal nuts.	Who held the nuts?
experimental	b	Mrs. Jones brought Peter to the museum. Immediately afterwards he bought a couple of postcards.	Who bought the postcards?
experimental	c	Peter helped Mr. Smith by the door. Immediately afterwards he bought a couple of oil cans.	Who bought the oil cans?
experimental	d	Peter helped Mrs. Jones by the house. Immediately afterwards he bought a couple of soap bars.	Who bought the soap bars?
experimental	a	Mr. Smith hugged Peter by the stove. Immediately afterwards he ate a couple of sandwiches.	Who ate the sandwiches?
experimental	b	Mrs. Jones scolded Peter by the fridge. Immediately afterwards he ate a couple of pancakes.	Who ate the pancakes?
experimental	c	Peter helped Mr. Smith in the office. Immediately afterwards he sent a couple of documents.	Who sent the documents?
experimental	d	Peter found Mrs. Jones by the path. Immediately afterwards he drank a couple of glasses of water.	Who drank the water?
experimental	a	Mr. Smith advised Peter by the tent. Immediately afterwards he bought a couple of oranges.	Who bought the oranges?
experimental	b	Mrs. Jones helped Peter by the shop. Immediately afterwards he chose a couple of rugby balls.	Who chose the rugby balls?
experimental	c	Peter scolded Mr. Smith by the sink. Immediately afterwards he cleaned a couple of forks.	Who cleaned the forks?
experimental	d	Peter found Mrs. Jones at the bus stop. Immediately afterwards he bought a couple of tickets.	Who bought the tickets?

Type	Condition	Sentence	Comprehension question
experimental	a	Mr. Smith thanked Peter by the factory. Immediately afterwards he fixed a couple of lightbulbs.	Who fixed the lightbulbs?
experimental	b	Mrs. Jones hugged Peter by the garden. Immediately afterwards he picked a couple of flowers.	Who picked the flowers?
experimental	b	Mrs. Jones hugged Peter by the tree. Immediately afterwards he picked a couple of green leaves.	Who picked the green leaves?
experimental	c	Peter helped Mr. Smith by the table. Immediately afterwards he ate a couple of nice steaks.	Who ate the steaks?
experimental	d	Peter thanked Mrs. Jones by the car. Immediately afterwards he held a couple of metal nuts.	Who held the nuts?
experimental	a	Mr. Smith brought Peter to the museum. Immediately afterwards he bought a couple of postcards.	Who bought the postcards?
experimental	b	Mrs. Jones helped Peter by the door. Immediately afterwards he bought a couple of oil cans.	Who bought the oil cans?
experimental	c	Peter helped Mr. Smith by the house. Immediately afterwards he bought a couple of soap bars.	Who bought the soap bars?
experimental	d	Peter hugged Mrs. Jones by the stove. Immediately afterwards he ate a couple of sandwiches.	Who ate the sandwiches?
experimental	a	Mr. Smith scolded Peter by the fridge. Immediately afterwards he ate a couple of pancakes.	Who ate the pancakes?
experimental	b	Mrs. Jones helped Peter in the office. Immediately afterwards he sent a couple of documents.	Who sent the documents?
experimental	c	Peter found Mr. Smith by the path. Immediately afterwards he drank a couple of glasses of water.	Who drank the water?

Type	Condition	Sentence	Comprehension question
experimental	d	Peter advised Mrs. Jones by the tent. Immediately afterwards he bought a couple of oranges.	Who bought the oranges?
experimental	a	Mr. Smith helped Peter by the shop. Immediately afterwards he chose a couple of rugby balls.	Who chose the rugby balls?
experimental	b	Mrs. Jones scolded Peter by the sink. Immediately afterwards he cleaned a couple of forks.	Who cleaned the forks?
experimental	c	Peter found Mr. Smith at the bus stop. Immediately afterwards he bought a couple of tickets.	Who bought the tickets?
experimental	d	Peter thanked Mrs. Jones by the factory. Immediately afterwards he fixed a couple of lightbulbs.	Who fixed the lightbulbs?
experimental	a	Mr. Smith hugged Peter by the garden. Immediately afterwards he picked a couple of flowers.	Who picked the flowers?
experimental	a	Mr. Smith hugged Peter by the tree. Immediately afterwards he picked a couple of green leaves.	Who picked the green leaves?
experimental	b	Mrs. Jones helped Peter by the table. Immediately afterwards he ate a couple of nice steaks.	Who ate the steaks?
experimental	c	Peter thanked Mr. Smith by the car. Immediately afterwards he held a couple of metal nuts.	Who held the nuts?
experimental	d	Peter brought Mrs. Jones to the museum. Immediately afterwards he bought a couple of postcards.	Who bought the postcards?
experimental	a	Mr. Smith helped Peter by the door. Immediately afterwards he bought a couple of oil cans.	Who bought the oil cans?
experimental	b	Mrs. Jones helped Peter by the house. Immediately afterwards he bought a couple of soap bars.	Who bought the soap bars?

Type	Condition	Sentence	Comprehension question
experimental	c	Peter hugged Mr. Smith by the stove. Immediately afterwards he ate a couple of sandwiches.	Who ate the sandwiches?
experimental	d	Peter scolded Mrs. Jones by the fridge. Immediately afterwards he ate a couple of pancakes.	Who ate the pancakes?
experimental	a	Mr. Smith helped Peter in the office. Immediately afterwards he sent a couple of documents.	Who sent the documents?
experimental	b	Mrs. Jones found Peter by the path. Immediately afterwards he drank a couple of glasses of water.	Who drank the water?
experimental	c	Peter advised Mr. Smith by the tent. Immediately afterwards he bought a couple of oranges.	Who bought the oranges?
experimental	d	Peter helped Mrs. Jones by the shop. Immediately afterwards he chose a couple of rugby balls.	Who chose the rugby balls?
experimental	a	Mr. Smith scolded Peter by the sink. Immediately afterwards he cleaned a couple of forks.	Who cleaned the forks?
experimental	b	Mrs. Jones found Peter at the bus stop. Immediately afterwards he bought a couple of tickets.	Who bought the tickets?
experimental	c	Peter thanked Mr. Smith by the factory. Immediately afterwards he fixed a couple of lightbulbs.	Who fixed the lightbulbs?
experimental	d	Peter hugged Mrs. Jones by the garden. Immediately afterwards he picked a couple of flowers.	Who picked the flowers?
filler	a	Before Peter talked to Susan by the kitchen, the kitchen was quite dirty.	What was dirty?
filler	b	The doctor saw the waiter working in the restaurant, it was a nice place.	Who was working?
filler	c	The restaurant was closing but Susan was happily talking to Mr. Smith.	Who was happy?

Type	Condition	Sentence	Comprehension question
filler	d	Peter was talking to the doctor by the fountain, the fountain was pretty.	What was pretty?
filler	e	The waiter was serving the food to Mrs. Jones by the garden, the garden was nice.	Who served the food?
filler	f	Mr. Smith was paying the plumber for fixing the pipes. The plumber was tired.	Who was tired?
filler	a	After Susan helped Mr. Smith by the shop, the shop was quite organized.	Where were they?
filler	b	The plumber helped the waiter carry the wheelbarrow, it was quite heavy.	What were they carrying?
filler	c	The wheelbarrow was very heavy but Mrs. Jones was helping Peter.	What was heavy?
filler	d	Susan was asking the waiter for some ice cold water; she was sitting at the restaurant.	Who asked for water?
filler	e	The lawyer talked to Peter by the office; the office was quite disorganized.	Where were they?
filler	f	Susan was talking to Mrs. Jones about the patio. Susan was very excited.	Who was excited?
filler	a	As Mr. Smith argued with Peter by the box, the box got quite wet.	Who was angry?
filler	b	The waiter talked to the lawyer sitting at the patio; it was a very nice patio.	Who was sitting at the patio?
filler	c	The patio was very dirty but Peter and Mr. Smith did not care at all.	What was dirty?
filler	d	The plumber was working for Mrs. Jones fixing the fountain; the fountain was broken.	What was broken?
filler	e	Mrs. Jones was talking about work to the lawyer by the hall; the hall was empty.	Where were they?
filler	f	Peter was fighting with Susan about the candy. Peter was angry.	Who was angry?
filler	a	While Mrs. Jones was waiting for Peter by the car, the car was running out of gas.	Who was waiting?

Type	Condition	Sentence	Comprehension question
filler	b	The lawyer informed the doctor in the office; the office was a mess.	What was a mess?
filler	c	The fountain's water was unsafe; Mrs. Jones and Susan were very unhappy about it.	What was unsafe?
filler	d	The lawyer was talking to Mrs. Jones by the building; the building was huge.	Where were they?
filler	e	Mr. Smith was happy at the waiter for the nice steak; the nice steak was delicious.	Who was happy?
filler	f	Mrs. Jones was paying attention to Peter talk about the garden. Peter was very passionate.	Who was talking?
practice	–	Mr. Smith kissed Mrs. Jones in the garden. Immediately afterwards he presented a beautiful ring.	What did he present?
practice	–	Peter informed the chef in the kitchen; the kitchen was a mess.	What was a mess?
practice	–	The lawyer was reimbursing the waiter for the stolen money. The waiter looked weary.	Who looked weary?

A.2 Spanish Experimental Sentences

The following table reports the Spanish sentences used in the Spanish baseline and Danish L2 Spanish experiments. The table includes filler, training, and experimental items, together with their condition labels and comprehension questions.

Table A.2.: Spanish sentences, conditions, and comprehension questions for the filler, training, and experimental materials.

Type	Condition	Sentence	Comprehension question
filler	a	Antes de que Pedro hablase con Susana en la cocina, la cocina estaba bastante sucia.	¿Qué estaba sucio?

Type	Condition	Sentence	Comprehension question
filler	b	El doctor vio al camarero trabajando en el restaurante, era un sitio bonito.	¿Quién estaba trabajando?
filler	c	El restaurante estaba cerrando pero Susana estaba hablando felizmente con Don Antonio.	¿Quién estaba contento?
filler	d	Pedro estaba hablando con el doctor al lado de la fuente, la fuente era bonita.	¿Qué era bonito?
filler	e	El camarero estaba sirviéndole la comida a Doña María en el jardín, el jardín era bonito.	¿Quién sirvió la comida?
filler	f	Don Antonio estaba pagando al fontanero por arreglar las tuberías. El fontanero estaba cansado.	¿Quién estaba cansado?
filler	a	Después de que Susana ayudó a Don Antonio en la tienda, la tienda estaba bastante organizada.	¿Dónde estaban ellos?
filler	b	El fontanero ayudó al camarero llevando la carretilla, era bastante pesada.	¿Qué estaban llevando?
filler	c	La carretilla era bastante pesada pero Doña María estaba ayudando a Pedro.	¿Qué era pesado?
filler	d	Susana le pidió al camarero un poco de agua con hielo, ella estaba sentada en el restaurante.	¿Quién pidió?
filler	e	El abogado habló con Pedro en la oficina, la oficina estaba bastante desorganizada.	¿Dónde estaban ellos?
filler	f	Susana estaba hablando con Doña María sobre el patio. Susana estaba bastante emocionada.	¿Quién estaba emocionada?
filler	a	Mientras Don Antonio discutía con Pedro sobre el paquete, el paquete se mojó bastante.	¿Quién estaba enfadado?
filler	b	El camarero habló con el abogado que estaba sentado en el patio, era un patio bonito.	¿Quién estaba sentado?
filler	c	El patio estaba muy sucio pero a Pedro y a Don Antonio les daba igual.	¿Qué estaba sucio?

Type	Condition	Sentence	Comprehension question
filler	d	El fontanero estaba trabajando para Doña María arreglando la fuente, la fuente estaba rota.	¿Qué estaba roto?
filler	e	Doña María estaba hablando del trabajo con el abogado en el pasillo, el pasillo estaba vacío.	¿Dónde estaban ellos?
filler	f	Pedro estaba peleando con Susana sobre los caramelos. Pedro estaba enfadado.	¿Quién estaba enfadado?
filler	a	Mientras Doña María estaba esperando a Pedro en el coche, el coche se estaba quedando sin gasolina.	¿Quién estaba esperando?
filler	b	El abogado informó al doctor en la oficina, la oficina era un desastre.	¿Qué era un desastre?
filler	c	El agua de la fuente no era segura, Doña María y Susana estaban tristes por ello.	¿Qué no era seguro?
filler	d	El abogado estaba hablando con Doña María al lado del edificio, el edificio era muy grande.	¿Dónde estaban ellos?
filler	e	Don Antonio estaba contento con el camarero por el rico filete, el filete era delicioso.	¿Quién estaba contento?
filler	f	Doña María estaba prestándole atención a Pedro hablar sobre el jardín. Pedro estaba muy apasionado.	¿Quién estaba hablando?
practice	–	Don Antonio besó a Doña María en el jardín. Inmediatamente después, él le dio un precioso anillo.	¿Quién dio el anillo?
practice	–	Pedro informó al chef en la cocina, la cocina era un desastre.	¿Qué era un desastre?
practice	–	El abogado estaba reembolsando al camarero por el dinero robado. El camarero se veía cansado.	¿Quién estaba cansado?
experimental	d	Pedro abrazó a Doña María al lado del árbol. Inmediatamente después, él cogió algunas de las hojas verdes.	¿Quién cogió las hojas verdes?

Type	Condition	Sentence	Comprehension question
experimental	a	Don Antonio ayudó a Pedro al lado de la mesa. Inmediatamente después, él comió algunos de los ricos filetes.	¿Quién comió el filete?
experimental	b	Doña María agradeció a Pedro al lado del coche. Inmediatamente después, él cogió algunas de las tuercas metálicas.	¿Quién cogió las tuercas?
experimental	c	Pedro llevó a Don Antonio al museo. Inmediatamente después, él compró algunas de las postales.	¿Quién compró las postales?
experimental	d	Pedro ayudó a Doña María al lado de la puerta. Inmediatamente después, él compró algunas latas de pintura.	¿Quién compró las latas de pintura?
experimental	a	Don Antonio ayudó a Pedro al lado de la casa. Inmediatamente después, él compró algunas barras de jabón.	¿Quién compró las barras de jabón?
experimental	b	Doña María abrazó a Pedro al lado de la cocina. Inmediatamente después, él se comió alguno de los sándwiches.	¿Quién se comió el sándwich?
experimental	c	Pedro regañó a Don Antonio al lado del frigorífico. Inmediatamente después, él se comió algunos de los crepes.	¿Quién se comió los crepes?
experimental	d	Pedro ayudó a Doña María en la oficina. Inmediatamente después, él mandó algunos de los documentos.	¿Quién envió los documentos?
experimental	a	Don Antonio se encontró a Pedro por el camino. Inmediatamente después, él bebió un poco de agua fría.	¿Quién se bebió el agua?
experimental	b	Doña María aconsejó a Pedro al lado de la tienda de campaña. Inmediatamente después, él compró algunas de las naranjas.	¿Quién compró las naranjas?
experimental	c	Pedro ayudó a Don Antonio en la tienda de deportes. Inmediatamente después, él compró algunas de las pelotas de rugby.	¿Quién compró las pelotas?
experimental	d	Pedro regañó a Don Antonio al lado del fregadero. Inmediatamente después, él limpió algunos de los cubiertos.	¿Quién limpió los cubiertos?

Type	Condition	Sentence	Comprehension question
experimental	a	Don Antonio se encontró a Pedro en el autobús. Inmediatamente después, él pagó algunos de los tickets.	¿Quién pagó los tickets?
experimental	b	Doña María agradeció a Pedro al lado de la fábrica. Inmediatamente después, él arregló algunas de las luces.	¿Quién arregló las luces?
experimental	c	Pedro abrazó a Doña María en el jardín. Inmediatamente después, él cogió algunas de las flores.	¿Quién cogió las flores?
experimental	c	Pedro abrazó a Don Antonio al lado del árbol. Inmediatamente después, él cogió algunas de las hojas verdes.	¿Quién cogió las hojas verdes?
experimental	d	Pedro ayudó a Doña María al lado de la mesa. Inmediatamente después, él comió algunos de los ricos filetes.	¿Quién comió el filete?
experimental	a	Don Antonio agradeció a Pedro al lado del coche. Inmediatamente después, él cogió algunas de las tuercas metálicas.	¿Quién cogió las tuercas?
experimental	b	Doña María llevó a Pedro al museo. Inmediatamente después, él compró algunas de las postales.	¿Quién compró las postales?
experimental	c	Pedro ayudó a Don Antonio al lado de la puerta. Inmediatamente después, él compró algunas latas de pintura.	¿Quién compró las latas de pintura?
experimental	d	Pedro ayudó a Doña María al lado de la casa. Inmediatamente después, él compró algunas barras de jabón.	¿Quién compró las barras de jabón?
experimental	a	Don Antonio abrazó a Pedro al lado de la cocina. Inmediatamente después, él se comió alguno de los sándwiches.	¿Quién se comió el sándwich?
experimental	b	Doña María regañó a Pedro al lado del frigorífico. Inmediatamente después, él se comió algunos de los crepes.	¿Quién se comió los crepes?
experimental	c	Pedro ayudó a Don Antonio en la oficina. Inmediatamente después, él mandó algunos de los documentos.	¿Quién envió los documentos?

Type	Condition	Sentence	Comprehension question
experimental	d	Pedro se encontró a Doña María por el camino. Inmediatamente después, él bebió un poco de agua fría.	¿Quién se bebió el agua?
experimental	a	Don Antonio aconsejó a Pedro al lado de la tienda de campaña. Inmediatamente después, él compró algunas de las naranjas.	¿Quién compró las naranjas?
experimental	b	Doña María ayudó a Pedro en la tienda de deportes. Inmediatamente después, él compró algunas de las pelotas de rugby.	¿Quién compró las pelotas?
experimental	c	Pedro regañó a Doña María al lado del fregadero. Inmediatamente después, él limpió algunos de los cubiertos.	¿Quién limpió los cubiertos?
experimental	d	Pedro se encontró a Doña María en el autobús. Inmediatamente después, él pagó algunos de los tickets.	¿Quién pagó los tickets?
experimental	a	Don Antonio agradeció a Pedro al lado de la fábrica. Inmediatamente después, él arregló algunas de las luces.	¿Quién arregló las luces?
experimental	b	Doña María abrazó a Pedro en el jardín. Inmediatamente después, él cogió algunas de las flores.	¿Quién cogió las flores?
experimental	b	Doña María abrazó a Pedro al lado del árbol. Inmediatamente después, él cogió algunas de las hojas verdes.	¿Quién cogió las hojas verdes?
experimental	c	Pedro ayudó a Don Antonio al lado de la mesa. Inmediatamente después, él comió algunos de los ricos filetes.	¿Quién comió el filete?
experimental	d	Pedro agradeció a Doña María al lado del coche. Inmediatamente después, él cogió algunas de las tuercas metálicas.	¿Quién cogió las tuercas?
experimental	a	Don Antonio llevó a Pedro al museo. Inmediatamente después, él compró algunas de las postales.	¿Quién compró las postales?
experimental	b	Doña María ayudó a Pedro al lado de la puerta. Inmediatamente después, él compró algunas latas de pintura.	¿Quién compró las latas de pintura?

Type	Condition	Sentence	Comprehension question
experimental	c	Pedro ayudó a Don Antonio al lado de la casa. Inmediatamente después, él compró algunas barras de jabón.	¿Quién compró las barras de jabón?
experimental	d	Pedro abrazó a Doña María al lado de la cocina. Inmediatamente después, él se comió alguno de los sándwiches.	¿Quién se comió el sándwich?
experimental	a	Don Antonio regañó a Pedro al lado del frigorífico. Inmediatamente después, él se comió algunos de los crepes.	¿Quién se comió los crepes?
experimental	b	Doña María ayudó a Pedro en la oficina. Inmediatamente después, él mandó algunos de los documentos.	¿Quién envió los documentos?
experimental	c	Pedro se encontró a Don Antonio por el camino. Inmediatamente después, él bebió un poco de agua fría.	¿Quién se bebió el agua?
experimental	d	Pedro aconsejó a Doña María al lado de la tienda de campaña. Inmediatamente después, él compró algunas de las naranjas.	¿Quién compró las naranjas?
experimental	a	Don Antonio ayudó a Pedro en la tienda de deportes. Inmediatamente después, él compró algunas de las pelotas de rugby.	¿Quién compró las pelotas?
experimental	b	Doña María regañó a Pedro al lado del fregadero. Inmediatamente después, él limpió algunos de los cubiertos.	¿Quién limpió los cubiertos?
experimental	c	Pedro se encontró a Don Antonio en el autobús. Inmediatamente después, él pagó algunos de los tickets.	¿Quién pagó los tickets?
experimental	d	Pedro agradeció a Doña María al lado de la fábrica. Inmediatamente después, él arregló algunas de las luces.	¿Quién arregló las luces?
experimental	a	Don Antonio abrazó a Pedro en el jardín. Inmediatamente después, él cogió algunas de las flores.	¿Quién cogió las flores?
experimental	a	Don Antonio abrazó a Pedro al lado del árbol. Inmediatamente después, él cogió algunas de las hojas verdes.	¿Quién cogió las hojas verdes?

Type	Condition	Sentence	Comprehension question
experimental	b	Doña María ayudó a Pedro al lado de la mesa. Inmediatamente después, él comió algunos de los ricos filetes.	¿Quién comió el filete?
experimental	c	Pedro agradeció a Don Antonio al lado del coche. Inmediatamente después, él cogió algunas de las tuercas metálicas.	¿Quién cogió las tuercas?
experimental	d	Pedro llevó a Doña María al museo. Inmediatamente después, él compró algunas de las postales.	¿Quién compró las postales?
experimental	a	Don Antonio ayudó a Pedro al lado de la puerta. Inmediatamente después, él compró algunas latas de pintura.	¿Quién compró las latas de pintura?
experimental	b	Doña María ayudó a Pedro al lado de la casa. Inmediatamente después, él compró algunas barras de jabón.	¿Quién compró las barras de jabón?
experimental	c	Pedro abrazó a Don Antonio al lado de la cocina. Inmediatamente después, él se comió alguno de los sándwiches.	¿Quién se comió el sándwich?
experimental	d	Pedro regañó a Doña María al lado del frigorífico. Inmediatamente después, él se comió algunos de los crepes.	¿Quién se comió los crepes?
experimental	a	Don Antonio ayudó a Pedro en la oficina. Inmediatamente después, él mandó algunos de los documentos.	¿Quién envió los documentos?
experimental	b	Doña María se encontró a Pedro por el camino. Inmediatamente después, él bebió un poco de agua fría.	¿Quién se bebió el agua?
experimental	c	Pedro aconsejó a Don Antonio al lado de la tienda de campaña. Inmediatamente después, él compró algunas de las naranjas.	¿Quién compró las naranjas?
experimental	d	Pedro ayudó a Doña María en la tienda de deportes. Inmediatamente después, él compró algunas de las pelotas de rugby.	¿Quién compró las pelotas?
experimental	a	Don Antonio regañó a Pedro al lado del fregadero. Inmediatamente después, él limpió algunos de los cubiertos.	¿Quién limpió los cubiertos?

Type	Condition	Sentence	Comprehension question
experimental	b	Doña María se encontró a Pedro en el autobús. Inmediatamente después, él pagó algunos de los tickets.	¿Quién pagó los tickets?
experimental	c	Pedro agradeció a Don Antonio al lado de la fábrica. Inmediatamente después, él arregló algunas de las luces.	¿Quién arregló las luces?
experimental	d	Pedro abrazó a Doña María en el jardín. Inmediatamente después, él cogió algunas de las flores.	¿Quién cogió las flores?

A.3 Danish Experimental Sentences

The following table reports the Danish experimental sentences used in the Danish baseline experiment. The table includes the condition labels and the corresponding comprehension questions.

Table A.3.: Danish experimental sentences, conditions, and comprehension questions.

Type	Condition	Sentence	Comprehension question
experimental	d	Peter krammede fr. Jones ved træet. Straks efter plukkede han nogle af de grønne blade.	Hvem plukkede de grønne blade?
experimental	a	Hr. Smith krammede Peter ved træet. Straks efter plukkede han nogle af de grønne blade.	Hvem plukkede de grønne blade?
experimental	b	Fr. Jones krammede Peter ved træet. Straks efter plukkede han nogle af de grønne blade.	Hvem plukkede de grønne blade?
experimental	c	Peter krammede hr. Smith ved træet. Straks efter plukkede han nogle af de grønne blade.	Hvem plukkede de grønne blade?
experimental	a	Hr. Smith hjalp Peter ved bordet. Straks efter spiste han nogle af de dejlige steaks.	Hvem spiste de dejlige steaks?

Type	Condition	Sentence	Comprehension question
experimental	b	Fr. Jones hjalp Peter ved bordet. Straks efter spiste han nogle af de dejlige steaks.	Hvem spiste de dejlige steaks?
experimental	c	Peter hjalp hr. Smith ved bordet. Straks efter spiste han nogle af de dejlige steaks.	Hvem spiste de dejlige steaks?
experimental	d	Peter hjalp fr. Jones ved bordet. Straks efter spiste han nogle af de dejlige steaks.	Hvem spiste de dejlige steaks?
experimental	b	Fr. Jones takkede Peter ved bilen. Straks efter holdt han nogle af møtrikkerne.	Hvem holdt møtrikkerne?
experimental	a	Hr. Smith takkede Peter ved bilen. Straks efter holdt han nogle af møtrikkerne.	Hvem holdt møtrikkerne?
experimental	c	Peter takkede hr. Smith ved bilen. Straks efter holdt han nogle af møtrikkerne.	Hvem holdt møtrikkerne?
experimental	d	Peter takkede fr. Jones ved bilen. Straks efter holdt han nogle af møtrikkerne.	Hvem holdt møtrikkerne?
experimental	c	Peter bar hr. Smith hen til museet. Straks efter købte han nogle af postkortene.	Hvem købte postkortene?
experimental	a	Hr. Smith bar Peter hen til museet. Straks efter købte han nogle af postkortene.	Hvem købte postkortene?
experimental	b	Fr. Jones bar Peter hen til museet. Straks efter købte han nogle af postkortene.	Hvem købte postkortene?
experimental	d	Peter bar fr. Jones hen til museet. Straks efter købte han nogle af postkortene.	Hvem købte postkortene?
experimental	d	Peter hjalp fr. Jones ved døren. Straks efter købte han nogle af oliekanderne.	Hvem købte oliekanderne?
experimental	a	Hr. Smith hjalp Peter ved døren. Straks efter købte han nogle af oliekanderne.	Hvem købte oliekanderne?
experimental	b	Fr. Jones hjalp Peter ved døren. Straks efter købte han nogle af oliekanderne.	Hvem købte oliekanderne?
experimental	c	Peter hjalp hr. Smith ved døren. Straks efter købte han nogle af oliekanderne.	Hvem købte oliekanderne?
experimental	a	Hr. Smith hjalp Peter ved huset. Straks efter købte han nogle stykker af sæberne.	Hvem købte sæberne?
experimental	b	Fr. Jones hjalp Peter ved huset. Straks efter købte han nogle stykker af sæberne.	Hvem købte sæberne?

Type	Condition	Sentence	Comprehension question
experimental	c	Peter hjalp hr. Smith ved huset. Straks efter købte han nogle stykker af sæberne.	Hvem købte sæberne?
experimental	d	Peter hjalp fr. Jones ved huset. Straks efter købte han nogle stykker af sæberne.	Hvem købte sæberne?
experimental	b	Fr. Jones krammede Peter ved komfuret. Straks efter spiste han nogle af sandwichene.	Hvem spiste sandwichene?
experimental	a	Hr. Smith krammede Peter ved komfuret. Straks efter spiste han nogle af sandwichene.	Hvem spiste sandwichene?
experimental	c	Peter krammede hr. Smith ved komfuret. Straks efter spiste han nogle af sandwichene.	Hvem spiste sandwichene?
experimental	d	Peter krammede fr. Jones ved komfuret. Straks efter spiste han nogle af sandwichene.	Hvem spiste sandwichene?
experimental	c	Peter skældte hr. Smith ud ved køleskabet. Straks efter spiste han nogle af pandekagerne.	Hvem spiste pandekagerne?
experimental	a	Hr. Smith skældte Peter ud ved køleskabet. Straks efter spiste han nogle af pandekagerne.	Hvem spiste pandekagerne?
experimental	b	Fr. Jones skældte Peter ud ved køleskabet. Straks efter spiste han nogle af pandekagerne.	Hvem spiste pandekagerne?
experimental	d	Peter skældte fr. Jones ud ved køleskabet. Straks efter spiste han nogle af pandekagerne.	Hvem spiste pandekagerne?
experimental	d	Peter hjalp fr. Jones på kontoret. Straks efter sendte han nogle af dokumenterne.	Hvem sendte dokumenterne?
experimental	a	Hr. Smith hjalp Peter på kontoret. Straks efter sendte han nogle af dokumenterne.	Hvem sendte dokumenterne?
experimental	b	Fr. Jones hjalp Peter på kontoret. Straks efter sendte han nogle af dokumenterne.	Hvem sendte dokumenterne?
experimental	c	Peter hjalp hr. Smith på kontoret. Straks efter sendte han nogle af dokumenterne.	Hvem sendte dokumenterne?

Type	Condition	Sentence	Comprehension question
experimental	a	Hr. Smith fandt Peter ved stien. Straks efter drak han noget af det kolde vand.	Hvem drak det kolde vand?
experimental	b	Fr. Jones fandt Peter ved stien. Straks efter drak han noget af det kolde vand.	Hvem drak det kolde vand?
experimental	c	Peter fandt hr. Smith ved stien. Straks efter drak han noget af det kolde vand.	Hvem drak det kolde vand?
experimental	d	Peter fandt fr. Jones ved stien. Straks efter drak han noget af det kolde vand.	Hvem drak det kolde vand?
experimental	b	Fr. Jones rådgav Peter ved teltet. Straks efter købte han nogle af appelsinerne.	Hvem købte appelsinerne?
experimental	a	Hr. Smith rådgav Peter ved teltet. Straks efter købte han nogle af appelsinerne.	Hvem købte appelsinerne?
experimental	c	Peter rådgav hr. Smith ved teltet. Straks efter købte han nogle af appelsinerne.	Hvem købte appelsinerne?
experimental	d	Peter rådgav fr. Jones ved teltet. Straks efter købte han nogle af appelsinerne.	Hvem købte appelsinerne?
experimental	c	Peter hjalp hr. Smith ved butikken. Straks efter valgte han nogle af de grimme rugbybolde.	Hvem valgte rugbyboldene?
experimental	a	Hr. Smith hjalp Peter ved butikken. Straks efter valgte han nogle af de grimme rugbybolde.	Hvem valgte rugbyboldene?
experimental	b	Fr. Jones hjalp Peter ved butikken. Straks efter valgte han nogle af de grimme rugbybolde.	Hvem valgte rugbyboldene?
experimental	d	Peter hjalp fr. Jones ved butikken. Straks efter valgte han nogle af de grimme rugbybolde.	Hvem valgte rugbyboldene?
experimental	c	Peter skældte fr. Jones ud ved vasken. Straks efter vaskede han noget af bestikket.	Hvem vaskede bestikket?
experimental	a	Hr. Smith skældte Peter ud ved vasken. Straks efter vaskede han noget af bestikket.	Hvem vaskede bestikket?
experimental	b	Fr. Jones skældte Peter ud ved vasken. Straks efter vaskede han noget af bestikket.	Hvem vaskede bestikket?
experimental	d	Peter skældte hr. Smith ud ved vasken. Straks efter vaskede han noget af bestikket.	Hvem vaskede bestikket?

Type	Condition	Sentence	Comprehension question
experimental	a	Hr. Smith fandt Peter i bussen. Straks efter betalte han nogle af billetterne.	Hvem betalte billetterne?
experimental	b	Fr. Jones fandt Peter i bussen. Straks efter betalte han nogle af billetterne.	Hvem betalte billetterne?
experimental	c	Peter fandt hr. Smith i bussen. Straks efter betalte han nogle af billetterne.	Hvem betalte billetterne?
experimental	d	Peter fandt fr. Jones i bussen. Straks efter betalte han nogle af billetterne.	Hvem betalte billetterne?
experimental	b	Fr. Jones takkede Peter ved fabrikken. Straks efter reparerede han nogle af lyspærerne.	Hvem reparerede lyspærerne?
experimental	a	Hr. Smith takkede Peter ved fabrikken. Straks efter reparerede han nogle af lyspærerne.	Hvem reparerede lyspærerne?
experimental	d	Peter takkede fr. Jones ved fabrikken. Straks efter reparerede han nogle af lyspærerne.	Hvem reparerede lyspærerne?
experimental	c	Peter takkede hr. Smith ved fabrikken. Straks efter reparerede han nogle af lyspærerne.	Hvem reparerede lyspærerne?
experimental	c	Peter krammede hr. Smith ved haven. Straks efter plukkede han nogle af blomsterne.	Hvem plukkede blomsterne?
experimental	b	Fr. Jones krammede Peter ved haven. Straks efter plukkede han nogle af blomsterne.	Hvem plukkede blomsterne?
experimental	a	Hr. Smith krammede Peter ved haven. Straks efter plukkede han nogle af blomsterne.	Hvem plukkede blomsterne?
experimental	d	Peter krammede fr. Jones ved haven. Straks efter plukkede han nogle af blomsterne.	Hvem plukkede blomsterne?
filler	a	Før Peter talte med Susan i køkkenet, var køkkenet ret beskidt.	Hvad var beskidt?
filler	b	Lægen så tjeneren arbejde i restauranten, det var et pænt sted.	Hvem arbejdede?
filler	c	Restauranten var ved at lukke, men Susan talte glad med hr. Smith.	Hvem var glad?
filler	d	Peter talte med lægen ved springvandet, springvandet var pænt.	Hvad var pænt?

Type	Condition	Sentence	Comprehension question
filler	e	Tjeneren serverede maden for fr. Jones i haven, haven var pæn.	Hvem serverede maden?
filler	f	Hr. Smith betalte blikkenslageren for at reparere rørene. Blikkenslageren var træt.	Hvem var træt?
filler	a	Efter Susan hjalp hr. Smith i butikken, var butikken ret organiseret.	Hvor var de?
filler	b	Blikkenslageren hjalp tjeneren med at bære trillebøren, den var ret tung.	Hvad bar de?
filler	c	Trillebøren var ret tung, men fr. Jones hjalp Peter.	Hvad var tungt?
filler	d	Susan bad tjeneren om noget iskoldt vand, hun sad i restauranten.	Hvem bad om vand?
filler	e	Advokaten talte med Peter på kontoret, kontoret var ret uorganiseret.	Hvor var de?
filler	f	Susan talte med fr. Jones om terrassen. Susan var ret begejstret.	Hvem var begejstret?
filler	a	Mens hr. Smith skændtes med Peter om pakken, blev pakken ret våd.	Hvem var vred?
filler	b	Tjeneren talte med advokaten, som sad på terrassen, det var en pæn terrasse.	Hvem sad på terrassen?
filler	c	Terrassen var meget beskidt, men Peter og hr. Smith var ligeglade.	Hvad var beskidt?
filler	d	Blikkenslageren arbejdede for fr. Jones med at reparere springvandet, springvandet var i stykker.	Hvad var i stykker?
filler	e	Fr. Jones talte om arbejde med advokaten i gangen, gangen var tom.	Hvor var de?
filler	f	Peter skændtes med Susan om slikket. Peter var vred.	Hvem var vred?
filler	a	Mens fr. Jones ventede på Peter ved bilen, var bilen ved at løbe tør for benzin.	Hvem ventede?
filler	b	Advokaten informerede lægen på kontoret, kontoret var et rod.	Hvad var et rod?
filler	c	Vandet i springvandet var ikke sikkert, fr. Jones og Susan var kede af det.	Hvad var ikke sikkert?

Type	Condition	Sentence	Comprehension question
filler	d	Advokaten talte med fr. Jones ved bygningen, bygningen var meget stor.	Hvor var de?
filler	e	Hr. Smith var glad for tjeneren på grund af den gode steak, steaken var lækker.	Hvem var glad?
filler	f	Fr. Jones lyttede til Peter, der talte om haven. Peter var meget passioneret.	Hvem talte?
practice	–	Hr. Smith kyssede fr. Jones i haven. Straks efter gav han hende en smuk ring.	Hvem gav ringen?
practice	–	Peter informerede kokken i køkkenet, køkkenet var et rod.	Hvad var et rod?
practice	–	Advokaten refunderede tjeneren de stjålne penge. Tjeneren så træt ud.	Hvem var træt?

A.4 Example Visual Stimuli

Because the full visual stimulus set was large, only representative examples are included in the appendix. The examples below illustrate the structure of the visual displays used in the visual-world experiment, including the placement of character regions, scenery/background regions, and comprehension questions. The full set consisted of character images, object images, and scenery/background images arranged into comparable displays across trials.

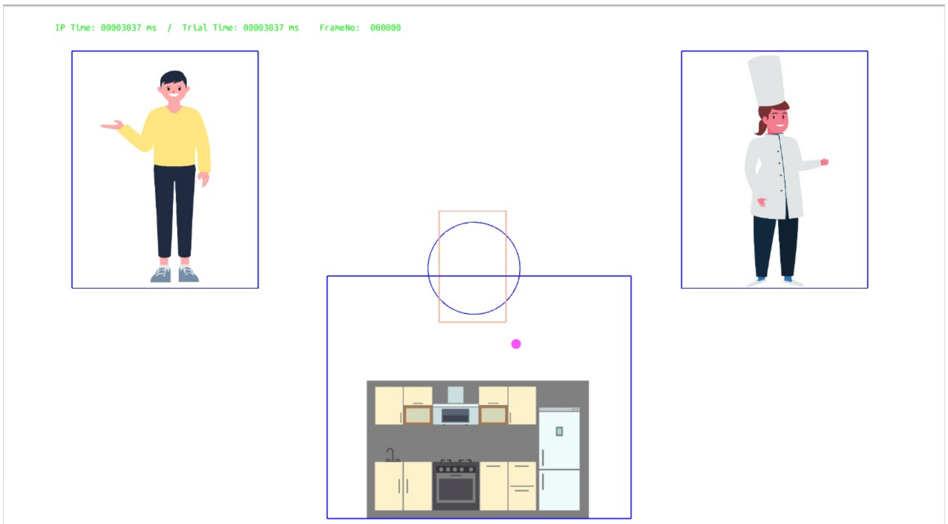

Figure A.1.: Example training display from the visual-world task. Training trials familiarized participants with the visual layout and task structure before the experimental trials began.

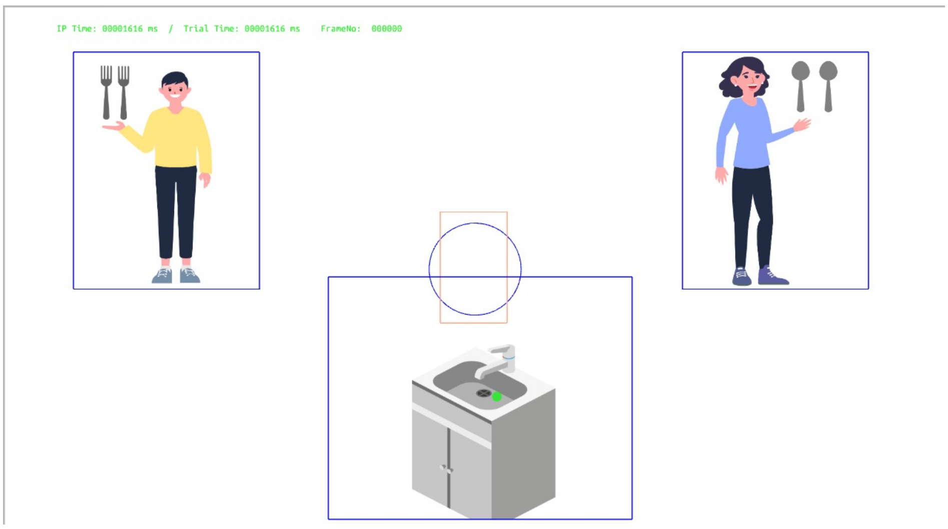

Figure A.2.: Example experimental display from the visual-world task with Peter and Mrs. Jones as the two potential antecedent characters. The blue boxes indicate the regions of interest used for gaze mapping.

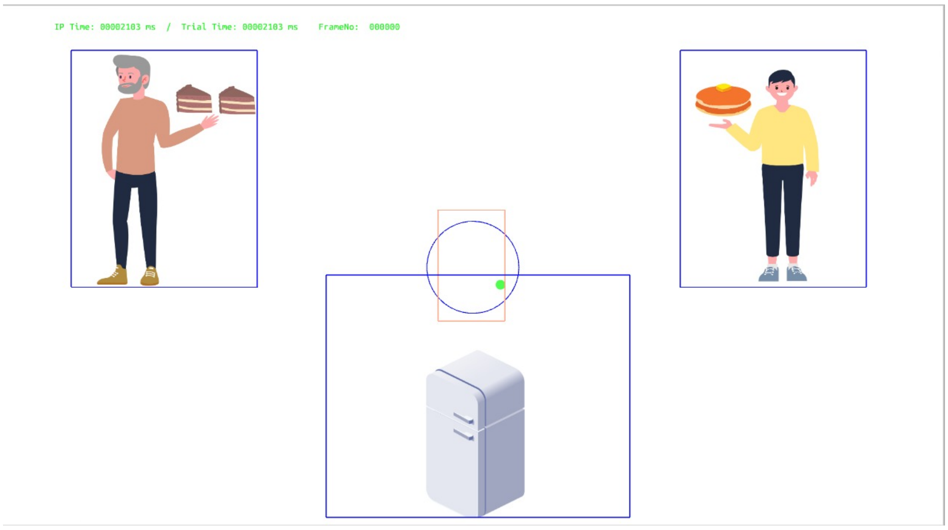

Figure A.3.: Example experimental display from the visual-world task with Peter and Mr. Smith as the two potential antecedent characters. The display shows two character regions and a scenery/background region.

Figure A.4.: Example comprehension-question screen. Participants answered by clicking on the image region corresponding to their interpretation of the ambiguous pronoun.